

# Designing Experiments to Predict Unit-Specific Treatment Effects: A Bayesian Bridge with Implementable Tuning

JoonHo Lee

**ABSTRACT.** Randomized experiments are typically designed to detect an average treatment effect, but their results are increasingly used to predict treatment effects for individual units. Tipton and Mamakos made such prediction an explicit design goal, evaluating sample size, moderator count, and site selection by the mean squared error of the predicted unit-specific effects. We develop a Bayesian decision-theoretic counterpart to their calculus and show that the two agree: under diffuse priors, the design-stage posterior-predictive criterion recovers their benchmark to first order, with exact finite-sample and cross-population corrections showing where the first-order arithmetic understates the risk. The Bayesian form makes several design levers explicit, including prior evidence from related studies, a continuum of analyses between the average-effect and fully moderated endpoints, the match between the experimental sample and the prediction target, and the correlation between potential outcomes, which no experiment can identify and which determines whether a stated accuracy target is attainable. The continuum raises a tuning problem that the design criterion cannot settle, because averaged under its own prior it always favors more shrinkage. Transferring classical shrinkage theory resolves it, with conditions attached: with three or more moderators, positive-part shrinkage protects against the moderated analysis’s estimation cost at a bounded price; no data-driven rule can protect both endpoints; and pretesting for moderation performs worst precisely where the moderation question is genuinely uncertain. A single matrix, computable at the design stage, prices the prediction burden of a mismatched target and decides whether a chosen tuning rule keeps its guarantee.

**Keywords:** experimental design; heterogeneous treatment effects; Bayesian decision theory; posterior-predictive risk; shrinkage estimation; empirical Bayes; generalizability

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Lee: The University of Alabama, Tuscaloosa, AL, USA. jlee296@ua.edu.

## 1. Introduction

Randomized experiments in education and the behavioral sciences are usually designed to detect an average treatment effect with adequate power (Bloom, 1995; Dong & Maynard, 2013). The questions that motivate these experiments, however, are increasingly unit specific. A district asks whether an online homework platform will help its own schools, a teacher asks whether a curriculum will help a particular student, and a policymaker asks where an intervention should be scaled first. Answering such questions requires predicting the treatment effect for individual units, and a design tuned for average-effect detection need not be a good design for unit-specific prediction.

Tipton and Mamakos (2025) reframed experimental design around this second goal. In their framework the target of inference is the unit-specific treatment effect (USTE), the design criterion is the mean squared prediction error (MSPE) of the predicted effects, and familiar design choices, including the sample size, the number of moderators, and the population from which the sample is drawn, are evaluated by how much they reduce prediction risk rather than by power alone. Their analysis yields a compact calculus. Within the experimental population, the MSPE of a moderator-based predictor decomposes into an estimation component, which grows with the number of moderators and shrinks with the arm sizes, plus a residual component that no model using the same covariates can remove. When the prediction target is a different population, the estimation component is scaled by an explicit factor built from the match between the sample and the target. Tipton and Mamakos develop the consequences in detail: the factor decomposes into a mean-shift term and a dispersion term, a source that is deliberately more heterogeneous than the target can drive the factor below its representative benchmark, and the residual component rests on a correlation between potential outcomes that no experiment can identify, which they handle by sensitivity analysis. The calculus builds on an established design tradition for generalizability and heterogeneity (e.g., Stuart et al., 2011; Tipton, 2013, 2014, 2021), and it gives applied researchers concrete planning quantities where previously there were only informal warnings.

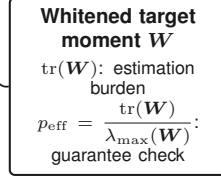
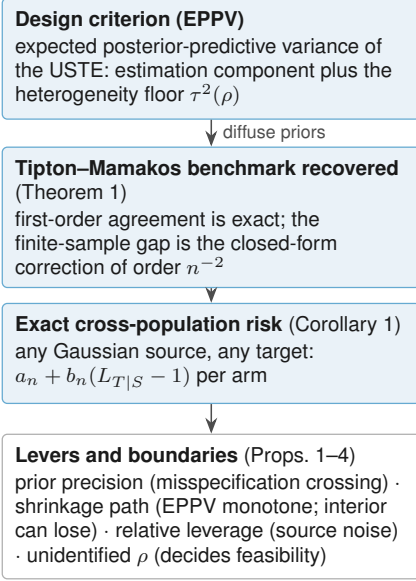
The calculus is frequentist, and in this article we ask what the same design arithmetic looks like on a Bayesian decision-theoretic foundation. The question is not ornamental, because several quantities a designer actually holds live naturally on that foundation. A designer often has prior evidence from related trials or multisite syntheses (Weiss et al., 2017), and the calculus has no channel through which such

evidence changes the arithmetic. The calculus also compares an average-effect predictor with an unregularized moderator predictor as two discrete options, whereas modern practice regularizes between the extremes (Hahn et al., 2020; Künzel et al., 2019). Once a shrinkage path connects the two endpoints, moreover, someone must choose a point on it, and it turns out that the design criterion itself cannot referee that choice.

The first half of the article establishes a bridge. We work in an arm-wise Gaussian linear working model with the residual variances and the cross-arm residual correlation held fixed, a setting we call the conditional layer, and we take the expected posterior-predictive variance (EPPV) of the USTE as the design criterion. Under diffuse priors, the conditional EPPV recovers the Tipton–Mamakos MSPE benchmark exactly to first order. The finite-sample gap between the two is an explicit inverse-moment correction of order one over the squared arm size, and a corollary extends the closed form to any Gaussian source paired with any target population, so the cross-population arithmetic no longer requires a first-order approximation. The bridge makes four design levers explicit: prior precision, a shrinkage continuum between the two endpoint analyses, relative leverage for a target population, and the unidentified correlation. For each lever we state the positive result together with the boundary at which it fails.

The second half turns to the tuning problem that the bridge exposes. Two observations motivate it. The oracle choice on the shrinkage path is weakly better than both endpoints, but it is an oracle; and the design-stage EPPV, evaluated under the prior that defines the path, decreases monotonically in the penalty, so it always asks for more shrinkage. What remains is the question practitioners actually face: which data-driven rules should tune the path, and what can be guaranteed about them. In a declared canonical regime the tuning problem reduces to a Gaussian mean problem, and classical shrinkage theory (Efron & Morris, 1973; James & Stein, 1961; Stein, 1981) answers the design question once it is transferred with its conditions attached. With three or more moderators, a positive-part James–Stein rule never predicts worse than the moderated endpoint, at a bounded premium when moderation is absent. No data-driven rule is weakly better than both endpoints across all truths, so this one-sided protection cannot be upgraded to a two-sided one. Stein’s unbiased risk estimate supplies the adjudication the prior cannot, because it targets the fixed-truth layer on which the endpoints disagree. For a mismatched target we specialize the classical quadratic-loss theory of Bock (1975) and Berger (1976) to

### The bridge (Sections 3–5)



### Implementable tuning (Section 6)

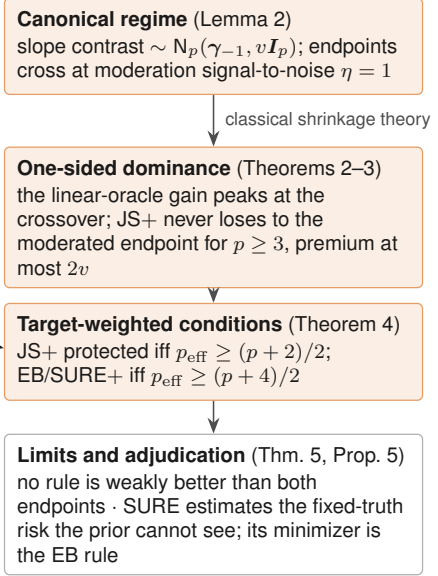


FIGURE 1. Overview of the article. The left column is the bridge half: the design-stage criterion recovers the Tipton–Mamakos benchmark to first order (Theorem 1), with an exact correction and an exact cross-population extension (Corollary 1), and exposes four design levers, each with a stated boundary. The right column is the tuning half: in the canonical regime, classical shrinkage theory yields one-sided guarantees (Theorem 3), their target-weighted conditions (Theorem 4), an impossibility bound (Theorem 5), and unbiased risk estimation (Proposition 5). The whitened target moment  $\mathbf{W}$  connects the halves: its trace prices the estimation burden of a mismatched target, and its effective dimension decides whether a chosen tuning rule keeps its guarantee.

the whitened target moment. The design contribution here is an identification: the matrix whose trace prices the target estimation burden also supplies the effective dimension and the rule-specific conditions that decide whether a chosen shrinkage rule keeps its guarantee. Figure 1 summarizes the structure of the article and the role of this matrix in connecting its two halves.

We also state what this article does not claim. We do not claim that Bayesian designs dominate frequentist designs, and no result below asserts an unconditional improvement. The bridge theorem has a dual frequentist reading, because under a flat prior the posterior covariance coincides with the conditional sampling covariance of ordinary least squares, so the finite-sample correction can be derived on either foundation; what the Bayesian formulation adds is the prior channel, the shrinkage

continuum, and the sensitivity vocabulary. We do not propose a new estimator of heterogeneous effects or a new general dominance theorem; every tuning rule and the Bock–Berger antecedent we use are classical, and our contribution is their target-design specialization, the identification of the relevant target matrix at the planning stage, and the transfer to the ATE-versus-CATE decision with exact rule-specific conditions. Flexible learners (e.g., Chipman et al., 2010; Hill, 2011; Wager & Athey, 2018) are complements rather than competitors to a design-stage risk criterion, and policy learning, in which predicted effects feed an assignment rule, stays downstream (Athey & Wager, 2021; Jia et al., 2025; Kitagawa & Tetenov, 2018). Sample-side design for a target population, that is, the choice of whom to enroll, is the subject of a companion article (Lee, 2026) that builds on the leverage objects developed here.

The plan of the article is as follows. In Section 2 we review the Tipton–Mamakos calculus and the neighboring literatures. In Section 3 we define the conditional Bayesian model and prove the bridge and its exact cross-population corollary. In Section 4 we develop the three estimation-side levers, and in Section 5 we treat the unidentified correlation. In Section 6 we develop implementable tuning in the canonical regime, including the target-weighted guarantee. Section 7 reports the simulation study, Section 8 collects the implications for design, and Section 9 concludes with limitations. Proofs, derivations, and a computational verification suite covering every formal claim appear in Appendices A–F.

## 2. The Design Calculus and Neighboring Work

**2.1. The Tipton–Mamakos benchmark.** We first restate the framework of Tipton and Mamakos (2025) in the notation used throughout. Units  $i$  are sampled from a population, each with potential outcomes  $Y_i(1)$  and  $Y_i(0)$  under treatment and control, a moderator vector  $\mathbf{x}_i \in \mathbb{R}^p$ , and the augmented vector  $\tilde{\mathbf{x}}_i = (1, \mathbf{x}_i^\top)^\top$  of dimension  $q = p + 1$ . The prediction target is the unit-specific treatment effect

$$\delta_i = Y_i(1) - Y_i(0), \tag{1}$$

which is never observed for any unit (Holland, 1986). Under a linear model for the potential outcomes, the USTE decomposes as  $\delta_i = \tilde{\mathbf{x}}_i^\top \boldsymbol{\gamma} + \nu_i$ , where  $\boldsymbol{\gamma}$  collects the average effect and the moderator slopes, and  $\nu_i$  is the residual effect heterogeneity that the chosen covariates do not explain. The variance of  $\nu_i$  is

$$\tau_{P|x}^2(\rho) = \sigma_{1|x}^2 + \sigma_{0|x}^2 - 2\rho \sigma_{1|x} \sigma_{0|x}, \tag{2}$$

where  $\sigma_{a|x}^2$  is the residual outcome variance in arm  $a$  and  $\rho$  is the correlation between the two residual potential outcomes. Because no unit reveals both potential outcomes,  $\rho$  is not identified by the experiment. Tipton and Mamakos (2025) treat it by sensitivity analysis, and we return to it in Section 5.

For an ordinary least squares moderator predictor trained on  $n_1$  treated and  $n_0$  control units sampled from the same population as the prediction target, Tipton and Mamakos (2025) show that the prediction risk is, to first order,

$$\text{MSPE} = \left( \frac{\sigma_{0|x}^2}{n_0} + \frac{\sigma_{1|x}^2}{n_1} \right) (1 + p) + \tau_{P|x}^2(\rho). \quad (3)$$

The first term is the estimation component: each additional moderator adds one unit to the factor  $1 + p = q$ , and each arm contributes its residual variance divided by its size. The second term is the residual component, a floor below which no predictor built from the same covariates can fall. A simple predictor that sets all moderator slopes to zero, which we call the ATE endpoint, avoids most of the estimation component but pays the full unexplained heterogeneity; the unregularized moderator predictor, the CATE endpoint, chases the explainable heterogeneity but pays the  $q/n$ -type estimation price. Which endpoint has lower risk depends on the moderation strength and the sample size, and Tipton and Mamakos (2025) give the crossover conditions.

When the prediction target is a population other than the source of the sample, Tipton and Mamakos (2025) show that the estimation component is further scaled by the factor  $1 + M_{T|S} + D_{T|S}$ , in which  $M_{T|S}$  is a squared Mahalanobis distance between the covariate means and  $D_{T|S}$  compares the covariate dispersions. Two of their conclusions are used repeatedly below. The factor is not a one-directional penalty, because a source deliberately more heterogeneous than the target can push it below its representative value; and weighting adjustments trade mismatch against a variance inflation priced by an effective sample size (Shimodaira, 2000; Steingrimsdottir et al., 2023). In their school-level application, a completed evaluation with eighty schools in one state would have needed roughly 113 schools, after reweighting, to match the accuracy of eighty schools drawn nationally, an inflation factor of 1.42. We restate this cross-population factor in Bayesian terms in Section 4.3 and make it exact in Corollary 1.

**2.2. Related literatures.** This article draws on four literatures, and it is useful to state what each supplies. First, Bayesian experimental design has long evaluated

designs by expected posterior quantities, from the preposterior tradition (Raiffa & Schlaifer, 1961) and the decision-theoretic synthesis of Chaloner and Verdinelli (1995) to sample-size determination through posterior-variance criteria (Wang & Gelfand, 2002) and modern computational treatments (Rainforth et al., 2024). The EPPV criterion below specializes this tradition to the USTE prediction objective; it is not offered as a new general criterion. The specialization is nevertheless where the work lies, because the USTE target couples two arms, a non-identified correlation, and a floor term in a way that generic design theory does not spell out.

Second, Bayesian causal inference supplies the model-based machinery for potential outcomes (Imbens & Rubin, 2015; Rubin, 1978) together with clear warnings about it: posterior spread reflects the prior and the working model, model-based credibility does not substitute for design-based credibility, and quantities such as the potential-outcome correlation are at best partially identified (Gustafson, 2015; Leavitt, 2023; Li et al., 2023). We take these warnings as constraints. All results below are conditional on a working residual covariance, the assignment mechanism is a randomized design throughout, and  $\rho$  is a sensitivity parameter rather than a quantity the posterior learns.

Third, shrinkage theory supplies the tuning half. The inadmissibility of the unregularized Gaussian mean in three or more dimensions (James & Stein, 1961), the positive-part improvement (Baranchik, 1970), the empirical-Bayes reading (Casella, 1985; Efron & Morris, 1973; Morris, 1983), Stein’s unbiased risk estimate (Donoho & Johnstone, 1995; Stein, 1981), and the dominance theory for general quadratic loss (Berger, 1976; Bock, 1975) are classical; textbook treatments appear in Lehmann and Casella (1998). Shrinkage has also been developed specifically for heterogeneous treatment effects, including Bayesian subgroup shrinkage (Henderson et al., 2016), hierarchical partial pooling as a guard against overinterpreting site-level heterogeneity (von Hippel & Schuetze, 2025), and SURE-based tuning in heteroscedastic hierarchical models (Xie et al., 2012). Our contribution here is not a new estimator or a new dominance theorem; it is the transfer of the classical guarantees to the ATE-versus-CATE design decision, the target-design identification of the matrix  $\mathbf{W}$ , and the impossibility result that bounds all tuning.

Fourth, the heterogeneous-treatment-effect literature supplies flexible learners (Chipman et al., 2010; Hahn et al., 2020; Hill, 2011; Künzel et al., 2019; Wager & Athey, 2018). These are analysis-stage tools. Our question is which design and which

tuning rule make the eventual prediction risk small, and the linear working model is what makes the design-stage algebra tractable.

### 3. A Conditional Bayesian Model and the Bridge

**3.1. Model, priors, and risk layers.** All results in this article hold under the following working frame, whose parts we cite by label.

**Assumption 1 (Conditional arm-wise Gaussian design frame).**

- (A1) Potential outcomes are well defined (SUTVA) and treatment  $W_i \in \{0, 1\}$  is assigned by a two-arm randomized design independent of the potential outcomes.
- (A2) The unit of randomization is the unit of prediction.
- (A3) Experimental units are drawn from a source population  $S$  and predictions are made for units drawn from a target population  $T$ , with moderators standardized in  $S$ ; until Section 4.3 we take  $T = S$ .
- (A4) In the source, potential outcomes follow the arm-wise Gaussian linear working model  $Y_i(a) = \tilde{\mathbf{x}}_i^\top \boldsymbol{\beta}_a + \varepsilon_i(a)$ , in which the residual pair  $\{\varepsilon_i(0), \varepsilon_i(1)\}$  is bivariate Gaussian given  $\mathbf{x}_i$  with variances  $\sigma_{0,S}^2, \sigma_{1,S}^2$ .
- (A5) In the target, the same coefficient vectors  $\boldsymbol{\beta}_0, \boldsymbol{\beta}_1$  govern the conditional means (no coefficient drift), and the target residual pair has variances  $\sigma_{0,T}^2, \sigma_{1,T}^2$  and correlation  $\rho_T$ , which define the floor  $\tau_{T|x}^2(\rho_T)$  through (2); when  $T = S$  we write  $\sigma_a^2$  and  $\rho$  for both.
- (A6) All residual parameters are held fixed at working values, so every risk statement below is conditional on this residual structure;  $\rho_T$  is a sensitivity parameter, not an estimand.
- (A7) The coefficient priors are arm-wise independent Gaussians specified through their precisions:  $\boldsymbol{\beta}_a$  has prior mean  $\mathbf{m}_a$  and prior precision  $\mathbf{K}_a/\sigma_{a,S}^2$  with  $\mathbf{K}_a \succeq \mathbf{0}$ , so  $\mathbf{K}_a = \mathbf{0}$  is the flat prior and singular  $\mathbf{K}_a$  gives a partially flat prior.
- (A8) Wherever a posterior quantity is written, the posterior is proper:  $\mathbf{S}_a + \mathbf{K}_a \succ \mathbf{0}$  almost surely, where  $\mathbf{S}_a = \tilde{\mathbf{X}}_a^\top \tilde{\mathbf{X}}_a$ . Under a flat prior this requires  $\mathbf{S}_a \succ \mathbf{0}$ , which holds almost surely when  $n_a \geq q$  and the covariates are continuously distributed; under a partially flat prior the data must span the null space of  $\mathbf{K}_a$ .



Three of these conditions deserve comment. Condition (A5) is the transport assumption that makes cross-population prediction meaningful, and it deliberately separates four objects that a single symbol would conflate: the source estimation noise  $\sigma_{a,S}^2$ , which drives the estimation component; the target residual floor  $\tau_{T|x}^2$ , which no source choice can touch; the target correlation  $\rho_T$ , which the data cannot identify; and coefficient drift, which we assume away and flag as a limitation. Condition (A6) is what we mean by the conditional layer. Condition (A8) is easy to overlook but consequential: without it, a flat prior with  $n_a < q$  leaves the posterior improper, and no statement below would be defined. Coupled priors across arms would introduce cross-arm posterior covariance terms; Appendix A records that form for completeness.

Because several distinct risk objects circulate in this area under similar names, Table 1 fixes the vocabulary. The frequentist MSPE of Tipton and Mamakos (2025) treats parameters as fixed and averages over sampling. The posterior-predictive variance (PPV) is an analysis-stage object; averaging it over the data the design will generate gives the EPPV, the design-stage criterion we use. Fixed-truth prediction MSE evaluates a procedure at a fixed parameter value, and we use it where a Bayesian criterion would be circular, in particular for diagnosing shrinkage and prior misspecification. The canonical slope-block risk of Section 6 is a fixed-truth object in a reduced coordinate system. Every formal statement below names its layer.

For a future unit with augmented covariates  $\tilde{\mathbf{x}}$  drawn from  $T$ , squared-error loss makes the posterior mean the Bayes predictor, and the design-stage criterion is

$$R_c(d \mid \Sigma_\varepsilon) = \mathbb{E}_{D|d} \mathbb{E}_{\tilde{\mathbf{x}} \sim T} \{ \text{Var}(\delta \mid D, \tilde{\mathbf{x}}, \Sigma_\varepsilon) \}, \quad (4)$$

where  $d$  collects the design choices and the subscript  $c$  marks the conditional layer.

**3.2. The bridge and its exact extension.** The starting point is the posterior-predictive variance of a single future unit.

**Lemma 1 (Conditional posterior-predictive variance).** Under Assumption 1, for a future unit with covariates  $\tilde{\mathbf{x}}$  from  $T$ ,

$$\text{Var}(\delta \mid D, \tilde{\mathbf{x}}, \Sigma_\varepsilon) = \sigma_{1,S}^2 \tilde{\mathbf{x}}^\top (\mathbf{S}_1 + \mathbf{K}_1)^{-1} \tilde{\mathbf{x}} + \sigma_{0,S}^2 \tilde{\mathbf{x}}^\top (\mathbf{S}_0 + \mathbf{K}_0)^{-1} \tilde{\mathbf{x}} + \tau_{T|x}^2(\rho_T), \quad (5)$$

and this quantity does not depend on the observed outcomes. The proof is given in Appendix A.

TABLE 1. Risk layers used in this article. Every formal statement names its layer.

| Object                                             | Definition                                                                                          | Role here                                                                         |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| MSPE                                               | MSE of prediction at fixed parameters, averaged over sampling (Tipton & Mamakos, 2025)              | Anchor benchmark                                                                  |
| Posterior-predictive variance (PPV)                | $\text{Var}(\delta \mid D, \tilde{\mathbf{x}}, \Sigma_\varepsilon)$ after the data arrive           | Analysis-stage object                                                             |
| Conditional EPPV, $R_c(d \mid \Sigma_\varepsilon)$ | PPV averaged over target draws and over the data the design will generate, residual structure fixed | Design criterion; target of Lemma 1, Theorem 1, Corollary 1, Propositions 1 and 3 |
| Full Bayes risk                                    | EPPV further integrated over $(\sigma_0^2, \sigma_1^2, \rho)$                                       | Outside this article; entangles the non-identified $\rho$                         |
| Fixed-truth prediction MSE                         | Risk of a procedure at a fixed parameter value                                                      | Diagnostic layer for shrinkage (Proposition 2) and prior misspecification         |
| Canonical slope-block risk                         | Fixed-truth risk of the moderator-slope block in units of $v = 2\sigma^2/n$ (Lemma 2)               | Layer of Theorems 2 through 5 and Proposition 5                                   |
| Finite realized-design risk                        | Fixed-truth USTE risk under a realized design with estimated scale                                  | Layer of the finite-design simulation track                                       |

Because the conditional PPV is free of the outcomes, the outer expectation in (4) runs over the design matrices alone, and substituting (5) gives

$$R_c(d \mid \Sigma_\varepsilon) = \sigma_{1,S}^2 \mathbb{E}[\text{tr}\{\mathbf{M}_T(\mathbf{S}_1 + \mathbf{K}_1)^{-1}\}] + \sigma_{0,S}^2 \mathbb{E}[\text{tr}\{\mathbf{M}_T(\mathbf{S}_0 + \mathbf{K}_0)^{-1}\}] + \tau_{T|x}^2(\rho_T), \quad (6)$$

where  $\mathbf{M}_T = \mathbb{E}_T(\tilde{\mathbf{x}}\tilde{\mathbf{x}}^\top)$  is the augmented second-moment matrix of the target. The bridge is the evaluation of (6) in the diffuse limit.

**Theorem 1 (Conditional bridge to the benchmark).** Let Assumption 1 hold with  $T = S$ , diffuse priors, and design rows drawn independently with  $\mathbf{x}_i \sim \mathbf{N}_p(\mathbf{0}, \Sigma_S)$  after whitening in  $S$ , so that  $\mathbf{M}_S = \mathbf{I}_q$ ; the trace criterion is invariant to this linear reparameterization, so the whitening is a normalization rather than a restriction. Then for  $n_a > p + 2$ ,

$$R_c(d \mid \Sigma_\varepsilon) = \sigma_0^2 \ell_p(n_0) + \sigma_1^2 \ell_p(n_1) + \tau_{S|x}^2(\rho), \quad \ell_p(n) = \frac{1}{n} + \frac{p(1 + 1/n)}{n - p - 2}, \quad (7)$$

exactly. Expanding  $\ell_p$  gives

$$R_c(d \mid \Sigma_\varepsilon) = \underbrace{\left( \frac{\sigma_0^2}{n_0} + \frac{\sigma_1^2}{n_1} \right) (1+p) + \tau_{S|x}^2(\rho)}_{\text{the benchmark (3)}} + \sum_{a \in \{0,1\}} \sigma_a^2 \frac{p(p+3)}{n_a(n_a-p-2)}, \quad (8)$$

so the diffuse-prior conditional EPPV recovers the Tipton–Mamakos MSPE benchmark exactly to first order, and the finite-sample gap is the correction on the right, of order  $n_0^{-2} + n_1^{-2}$ , strictly positive whenever  $p \geq 1$ . The proof is given in Appendix A.

Theorem 1 has a dual reading. Under a flat prior with known residual variance, the posterior covariance  $\sigma_a^2 \mathbf{S}_a^{-1}$  equals the conditional sampling covariance of the ordinary least squares coefficients, so averaging over Gaussian designs produces the same correction on a frequentist fixed-truth layer. The theorem is therefore both a Bayesian bridge and an exact audit of the population-moment approximation in Tipton and Mamakos (2025): the first-order benchmark is confirmed, and its finite-sample error now has a closed form. The correction is not ignorable at realistic sizes. In the default two-moderator setting with unit residual variances and balanced arms it specializes to  $\Delta_n = 20/\{n(n-4)\}$ . At  $n = 20$  units per arm the exact risk exceeds the first-order benchmark by 0.0625 in outcome-variance units, about a fifth of the entire estimation component at that size, and by  $n = 100$  the correction has fallen below 0.003. Figure 2 displays the bridge together with Monte Carlo design averages that match the closed form; the log-log slope of the correction is  $-2.06$  (Appendix F).

The exact arithmetic is not confined to the representative case.

**Corollary 1 (Exact cross-population risk; exact Gaussian-design layer).**

Under the conditions of Theorem 1 except that predictions are for an arbitrary target with augmented second moment  $\mathbf{M}_T$  (source-standardized,  $\mathbf{M}_T[1,1] = 1$ ), and with source rows  $\mathbf{x}_i \sim \mathbf{N}_p(\boldsymbol{\mu}_S, \boldsymbol{\Sigma}_S)$ ,

$$\mathbb{E}[\text{tr}(\mathbf{M}_T \mathbf{S}_a^{-1})] = a_n + b_n (L_{T|S} - 1), \quad a_n = \frac{1}{n_a} + \frac{p}{n_a(n_a-p-2)}, \quad b_n = \frac{1}{n_a-p-2}, \quad (9)$$

where  $L_{T|S} = \text{tr}(\mathbf{M}_S^{-1} \mathbf{M}_T)$  is the relative leverage of Section 4.3. Consequently the exact conditional EPPV for any Gaussian source and any target is  $\sum_a \sigma_{a,S}^2 \{a_n + b_n (L_{T|S} - 1)\} + \tau_{T|x}^2(\rho_T)$ . The proof, an affine-equivariance argument that reduces every source to the standard one, is given in Appendix A.

Corollary 1 serves two purposes. It upgrades the cross-population lever of Section 4.3 from a first-order statement to an exact one, and it allows every figure that displays a cross-population risk to carry the exact value alongside the first-order line, so that the size of the first-order gap can be read off rather than taken on trust (Figures 4 and 5).

The corollary also introduces the object that later reconnects design to analysis. After source whitening, let  $\mathbf{W}$  denote the  $p$ -dimensional slope block of the target second moment. Then  $\text{tr}(\mathbf{W}) = L_{T|S} - 1$  prices the slope-estimation burden in Corollary 1, while  $p_{\text{eff}} = \text{tr}(\mathbf{W})/\lambda_{\max}(\mathbf{W})$  records how evenly that burden is spread across directions. Section 6.5 specializes the classical Bock–Berger theory to this target-design object: the trace prices prediction, and the spectrum determines the rule-specific dominance window. We therefore carry both summaries forward rather than reducing target mismatch to leverage alone.

#### 4. Design Levers on the Estimation Component

The bridge preserves the benchmark; its value lies in the levers it makes explicit. This section develops three levers that act on the estimation component. For each we state the positive result and the boundary at which the lever fails.

4.1. **Prior precision.** The first lever is prior information.

**Proposition 1 (Prior precision and the estimation component; conditional EPPV layer).** Under Assumption 1, suppose additionally that the diffuse comparator is proper, equivalently  $\mathbf{S}_a \succ \mathbf{0}$  for each realized arm design. For any prior precisions  $\mathbf{K}_a \succeq \mathbf{0}$  and any target second moment  $\mathbf{M}_T \succeq \mathbf{0}$ , the estimation component of  $R_c(d \mid \Sigma_\varepsilon)$  in (6) is less than or equal to its diffuse-prior value, with strict inequality when  $\mathbf{K}_a \succ \mathbf{0}$  and  $\mathbf{M}_T \succ \mathbf{0}$ . An informative prior may make  $\mathbf{S}_a + \mathbf{K}_a$  invertible when  $\mathbf{S}_a$  is singular, but then the diffuse comparison itself is undefined. The proof, a Loewner-ordering argument, is given in Appendix B.

Proposition 1 concerns posterior spread, not prediction error against a fixed truth, and the distinction matters because prior-mean error trades the variance gain for bias. In the verified two-moderator illustration, a half-residual-standard-deviation offset in the prior mean crosses above the diffuse MSE between  $n_0 = 20$  and 35. Under the restricted information-scaled specialization, balanced homoscedastic arms give the heuristic first-order component  $2\sigma^2 q/(n + n_0)$ . The crossing proof, the full numerical path, the restrictions, and the display are collected in Appendix B.

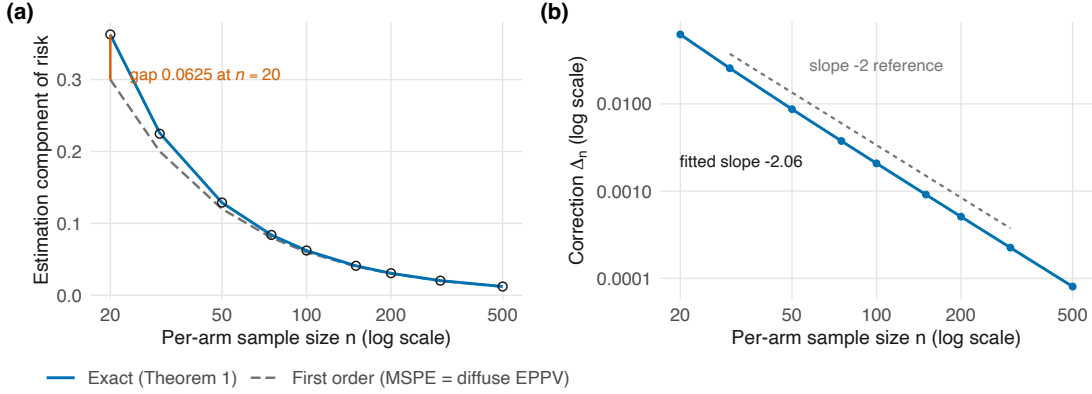


FIGURE 2. The conditional bridge in the default setting ( $p = 2$ ,  $\sigma_0 = \sigma_1 = 1$ ,  $\rho = 0$ , balanced arms). (a) The first-order risk, which is identical for the Tipton–Mamakos MSPE and the diffuse conditional EPPV, together with the exact finite-sample risk from Theorem 1 and Monte Carlo design averages. (b) The exact correction  $\Delta_n = 20/\{n(n-4)\}$  on log-log axes with a slope  $-2$  reference line.

**4.2. A shrinkage path between the endpoints.** The second lever replaces a discrete model choice with a continuum. Write the pooled interaction regression

$$Y_i = \tilde{\mathbf{x}}_i^\top \beta_0 + W_i \tilde{\mathbf{x}}_i^\top \gamma + \varepsilon_i, \quad (10)$$

in which  $\gamma$  carries the average effect in its first coordinate and the moderator slopes in  $\gamma_{-1}$ , and consider estimators  $\hat{\gamma}(\lambda)$  that penalize the moderator slopes by a ridge penalty  $\lambda$  while leaving the average-effect coordinate and the control regression unpenalized. For this construction we adopt a common residual scale  $\sigma$  across arms, because under that convention the penalized fit coincides exactly with a posterior mean under a mean-zero Gaussian slab on the slopes. The family connects the CATE endpoint ( $\lambda = 0$ ) to the covariate-adjusted ATE endpoint ( $\lambda \rightarrow \infty$ ), and it requires no observed unit-level effect contrast; details are given in Appendix C.

**Proposition 2 (Oracle shrinkage frontier; fixed-truth layer).** Fix a design  $d$  and a truth  $\gamma$ , and let  $R_\gamma(\lambda; d)$  denote the fixed-truth conditional prediction risk of  $\hat{\gamma}(\lambda)$  for the USTE, including the floor  $\tau^2$ . Then

$$\inf_{\lambda \in [0, \infty]} R_\gamma(\lambda; d) \leq \min\{R_\gamma(0; d), R_\gamma(\infty; d)\}, \quad (11)$$

because the endpoints belong to the closure of the path. The statement is an oracle statement: it asserts nothing about any data-driven rule for choosing  $\lambda$ .

Proposition 2 is elementary, but two computational observations attached to it carry the practical content and motivate Section 6. Figure 3 shows the fixed-truth risk along the path for three moderation-signal scenarios and three sample sizes; Appendix C reports the complete boundary table.

The first observation is that an interior choice can lose to the better endpoint. Across our grid of nine scenario-size cells with eight interior penalty values each, 36 of the 72 interior cells have higher fixed-truth risk than the better endpoint, with excess risk up to 0.22 in outcome-variance units, and in the no-signal scenario every interior  $\lambda$  is strictly worse than the ATE endpoint. The continuum widens the design vocabulary, but it does not by itself guarantee that some shrinkage helps. In the figure the best grid value is reported as a grid oracle, because it is selected from the ten penalty values evaluated rather than by continuous optimization.

The second observation is structural: the design-stage EPPV cannot locate the oracle. When the shrinkage penalty is read as its matching prior, the EPPV computed under that same prior decreases monotonically in  $\lambda$  in every cell of our grid, because a criterion evaluated under the path’s own prior always favors more shrinkage (Appendix C proves the monotonicity for every realized design). The frontier in (11) lives on the fixed-truth layer, where the truth can disagree with the prior. Adjudicating between the endpoints therefore requires either external information about moderation strength or an estimate of fixed-truth risk itself. Section 6 shows that the second route is available and comes with guarantees.

**4.3. Relative leverage for a target population.** The third lever concerns where the sample comes from. Suppose the design samples from  $S$  but predictions are needed for  $T \neq S$ .

**Proposition 3 (Relative leverage; exact Gaussian-design and first-order layers).** Under Assumption 1 with diffuse priors and Gaussian source rows, the estimation component of the target risk is exactly  $\sum_a \sigma_{a,S}^2 \{a_n + b_n(L_{T|S} - 1)\}$  by Corollary 1, and to first order it is

$$\left( \frac{\sigma_{0,S}^2}{n_0} + \frac{\sigma_{1,S}^2}{n_1} \right) L_{T|S}, \quad L_{T|S} = \text{tr}(\mathbf{M}_S^{-1} \mathbf{M}_T) = 1 + M_{T|S} + D_{T|S}, \quad (12)$$

matching the mean-shift and dispersion decomposition of Tipton and Mamakos (2025). Under  $S$ -standardization the representative benchmark is  $L_{T|S} = q$ , and  $L_{T|S}$  can fall below, equal, or exceed  $q$ . For non-Gaussian source rows the first-order

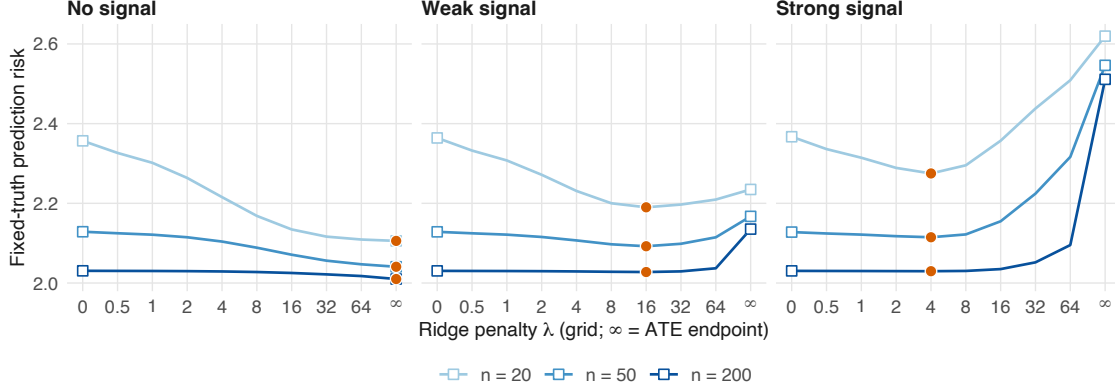


FIGURE 3. Fixed-truth prediction risk along the shrinkage path for three moderation-signal scenarios (columns) and three per-arm sizes. Open and filled squares mark the CATE endpoint ( $\lambda = 0$ ) and the ATE endpoint ( $\lambda = \infty$ ); circles mark the grid oracle, which is interior under weak and strong signal and sits at the ATE endpoint under no signal. Interior choices can exceed the better endpoint. The complete cell table is given in Appendix C.

statement holds under the regularity conditions stated in Appendix C, which include almost-sure nonsingularity and uniform integrability of the inverse trace; the Gaussian frame satisfies them.

We call  $L_{T|S}$  a relative leverage factor rather than a penalty, because Tipton and Mamakos (2025) established that mismatch is not one-directional, and the factor makes their observation computable scenario by scenario. Figure 4(a) computes  $L_{T|S}$  in five scenarios with  $q = 3$ . A source that is overdispersed relative to the target has leverage 1.50, half the representative value, because a wider design spreads the same estimation variance over a wider covariate range than the target requires; a source that is narrow in one moderator direction has leverage 11.25; and a mean-shifted target gives 5.50. No bound of the form  $L_{T|S} \geq q$  holds.

Low leverage does not by itself guarantee low risk, however, because leverage multiplies the residual variances of the source, and condition (A5) states the decoupling explicitly: the source outcome noise  $\sigma_{a,S}^2$  can differ from the target heterogeneity that sets the floor. Figure 4(b) evaluates the total conditional risk at  $n = 20$  per arm, showing both the first-order values and the exact values from Corollary 1. The overdispersed source with clean outcomes beats the representative design (2.15 against 2.30 first order; 2.18 against 2.36 exactly), but the same overdispersed source with  $\sigma_{a,S}^2 = 4$  loses (2.60 against 2.30 first order; 2.70 against 2.36 exactly) even though its

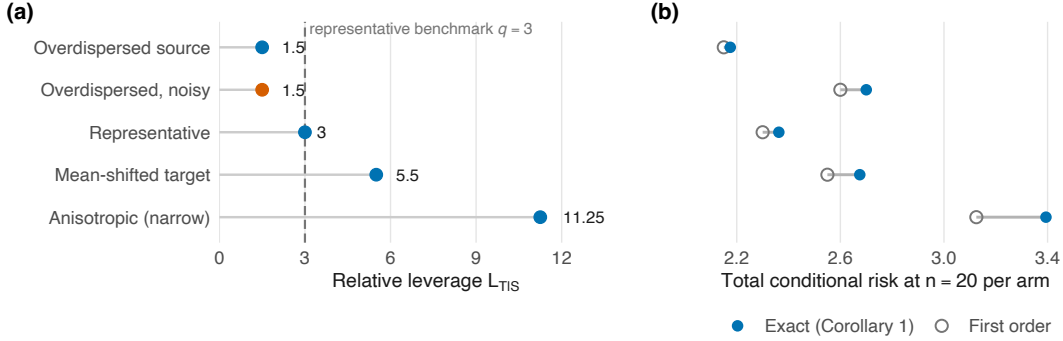


FIGURE 4. Relative leverage in five source-target scenarios ( $q = 3$ ;  $\rho_T = 0$ ). (a)  $L_{T|S}$  with the representative benchmark  $q = 3$  marked by the dashed line. (b) Total conditional risk at  $n = 20$  per arm: first-order values and exact values from Corollary 1. The noisy overdispersed source ties for the lowest leverage in panel (a) yet carries the second-highest exact risk in panel (b), because leverage multiplies the source residual variance  $\sigma_{a,S}^2 = 4$ .

leverage is still 1.50. Site selection for prediction therefore trades covariate coverage against outcome noise. The exact values also show that the first-order gap is largest where leverage is largest; the anisotropic scenario reads 3.13 first order against 3.39 exactly at  $n = 20$ , which is one more reason to plan with Corollary 1 at small sizes. Turning this lever into a sample-design procedure is the subject of the companion article (Lee, 2026).

## 5. Sensitivity to the Unidentified Correlation

The floor  $\tau_{T|x}^2(\rho_T)$  depends on the correlation between a unit's two residual potential outcomes. Tipton and Mamakos (2025) identify this dependence, call the correlation unknowable, and recommend sensitivity analysis. In this section we examine which design-stage conclusions the sensitivity does and does not affect.

**Remark 1 (The observed-data likelihood is free of  $\rho$ ).** Under Assumption 1, the observed-data likelihood factors into the two arms and contains  $\sigma_{0,S}^2$  and  $\sigma_{1,S}^2$  but not  $\rho$ . Under a prior in which  $\rho$  is a priori independent of the remaining parameters, any posterior statement about  $\rho$  reproduces its prior, so treating  $\rho$  as learned would be circular. The formal argument is given in Appendix D. This is the classical partial-identification problem (Aronow et al., 2014; Ding et al., 2019; Fan & Park, 2010; Heckman et al., 1997; Holland, 1986), and the Bayesian formulation does not escape it; its contribution is to force the dependence into the open (Gustafson, 2015).



**Proposition 4 (Restricted ranking invariance; conditional EPPV layer).** Fix the target, the model, and the marginal residual scales, and let candidate designs  $d \in \mathcal{D}$  differ only in their estimation components  $E(d)$ , so that  $R_c(d; \rho) = E(d) + \tau_{T|x}^2(\rho)$  with a common floor as  $\rho$  varies. Then the ranking of designs by  $R_c$  is invariant to  $\rho$ . The proof is immediate, and the restriction does the work: the invariance fails when candidate designs change the covariate set, the model, or the target, because the floor is then no longer common.

**Corollary 2 (Feasibility is  $\rho$ -sensitive).** A target prediction risk  $R^*$  is attainable at some finite sample size only if  $R^* > \tau_{T|x}^2(\rho)$ . In the balanced diffuse first-order setting the required per-arm size is

$$n^*(\rho) = \frac{2\sigma^2 q}{R^* - \tau_{T|x}^2(\rho)}, \quad (13)$$

a real-valued threshold whose implementable version is the smallest integer above it subject to  $n > p + 2$ , and  $n^*(\rho) = \infty$  when the margin is nonpositive. The exact-layer requirement solves  $2\sigma^2 \ell_p(n) = R^* - \tau^2$ , and it always exceeds the first-order requirement because the correction in Theorem 1 is positive for  $p \geq 1$ . Proofs appear in Appendix D.

Table 2 evaluates (13) in the default setting for the target  $R^* = 2.25$ , and Figure 5(b) traces both the first-order and the exact curves. At  $\rho = 0.3$  the first-order requirement is eight units per arm (the smallest integer above 7.06) and the exact requirement is eleven; at  $\rho = 0$  the pair is twenty-four against twenty-eight; and at  $\rho = -0.5$  the floor is 3.0, the margin is negative, and no sample size attains the target. The same experiment, the same model, and the same design options are therefore feasible or infeasible depending on a parameter that the data cannot identify. In our view this is the most consequential planning implication of the formulation, and it suggests that prediction-oriented power analyses should report feasibility over a stated range of  $\rho$ . Note also that the two corrections compound in the same direction, because the finite-sample gap in Theorem 1 is largest precisely where the first-order formula suggests that small samples suffice.

Because prediction intervals are the applied deliverable, we also verified their coverage in 4,000 replications per cell. Coverage is .950–.960 when the working correlation is correct. With true  $\rho = 0.3$  and assumed  $\rho = 0.9$ , it falls to .713 and .592 at twenty and one hundred units per arm, whereas assumptions below the truth err conservatively. Appendix D gives the protocol, the complete results, and the display.

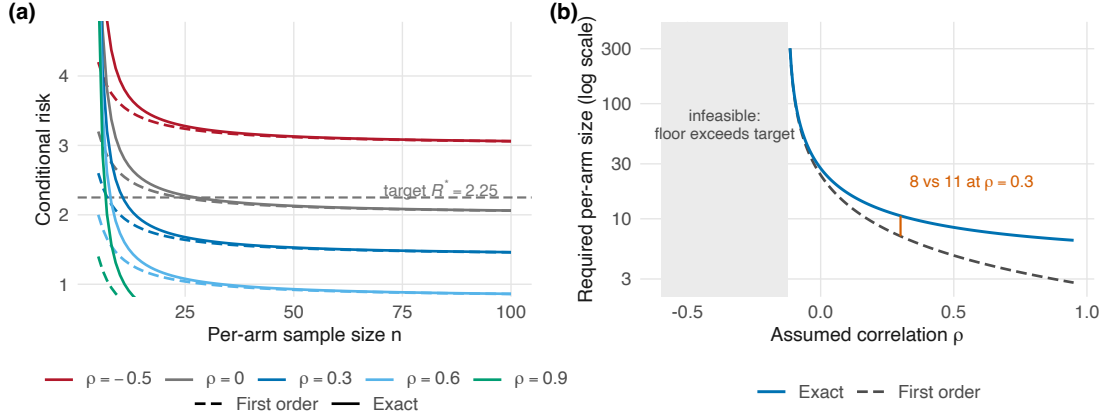


FIGURE 5. Sensitivity of the design calculus to the unidentified correlation ( $p = 2$ ,  $\sigma_0 = \sigma_1 = 1$ , balanced arms, diffuse priors). (a) Conditional risk against per-arm size for five values of  $\rho$ , first-order and exact, with the planning target  $R^* = 2.25$  marked; the  $\rho = -0.5$  curves never reach the target. (b) Required per-arm size on the first-order and exact layers; the shaded region marks assumed correlations for which the target is infeasible.

TABLE 2. Feasibility of the planning target  $R^* = 2.25$  in the default setting ( $p = 2$ ,  $\sigma_0 = \sigma_1 = 1$ , balanced arms, diffuse priors). The floor is  $\tau^2(\rho) = 2(1 - \rho)$  and the margin is  $R^* - \tau^2(\rho)$ . The first-order requirement is Corollary 2; the exact requirement solves the finite-sample form of Theorem 1 and always exceeds it. Implementable sizes are the smallest integers above the thresholds subject to  $n > p + 2$ .

| $\rho$ | Floor | Margin | Feasible | $n^*$ first order | $n^*$ exact |
|--------|-------|--------|----------|-------------------|-------------|
| -0.5   | 3.00  | -0.75  | no       | $\infty$          | $\infty$    |
| 0.0    | 2.00  | 0.25   | yes      | 24.0              | 27.4        |
| 0.3    | 1.40  | 0.85   | yes      | 7.1               | 10.6        |
| 0.6    | 0.80  | 1.45   | yes      | 4.1               | 7.8         |
| 0.9    | 0.20  | 2.05   | yes      | 2.9               | 6.6         |

## 6. Implementable Tuning in the Canonical Regime

The two observations of Section 4.2 rule out trusting the prior and picking a conventional penalty. Current practice resolves the endpoint choice a third way: analysts test for moderation and let the test pick the endpoint. The risk behavior of such pretest procedures is well understood in the model-selection literature, which predicts a risk bump near the detection boundary (Leeb & Pötscher, 2005), and our simulation study

places this stylized practice default on the same risk map as the tuned rules. This section develops what data-driven tuning can and cannot guarantee.

**6.1. The canonical regime.** All tuning theory in this article is stated in the following regime, and the simulation study quantifies its distance to realized designs.

**Lemma 2 (Canonical regime).** Let Assumption 1 hold with  $T = S$ , matched target moments ( $\mathbb{E}_T \mathbf{x} = \mathbf{0}$ ,  $\mathbb{E}_T \mathbf{x} \mathbf{x}^\top = \mathbf{I}_p$  after source whitening), a balanced two-arm design of  $n$  units per arm, common known residual scale  $\sigma$ , and the interaction regression (10). Then, conditional on the realized design, the OLS moderator-slope contrast is exactly Gaussian with a design-dependent covariance, and as  $n$  grows,

$$\sqrt{n}(\hat{\gamma}_{-1} - \gamma_{-1}) \Rightarrow \mathbf{N}_p(\mathbf{0}, 2\sigma^2 \mathbf{I}_p), \quad (14)$$

so to first order in the design average  $\hat{\gamma}_{-1} \sim \mathbf{N}_p(\gamma_{-1}, v \mathbf{I}_p)$  with  $v = 2\sigma^2/n$ . Under the matched target moments, the slope-block fixed-truth USTE risk of any predictor that replaces  $\hat{\gamma}_{-1}$  by  $\tilde{\gamma}_{-1}$  is  $\mathbb{E}\|\tilde{\gamma}_{-1} - \gamma_{-1}\|^2$ , up to the floor and average-effect terms that are common to every rule on the path. The derivation is given in Appendix E.

Each condition in Lemma 2 has a specific role. Matched target moments make the slope loss Euclidean; a mismatched target induces a weighted loss, which is the case Theorem 4 addresses. The exact conditional normality is a finite-sample statement, but the unconditional spherical form is asymptotic, because averaging over random designs mixes covariances; the finite-design track in Section 7 prices the difference. The known scale is a planning convention, and the simulation estimates it. In this regime a ridge penalty  $\lambda$  acts as a scalar factor  $c \in [0, 1]$  on the slope block, risks depend on the truth only through the signal size  $G = \|\gamma_{-1}\|^2$ , and the single scalar

$$\eta = \frac{G}{pv} \quad (15)$$

indexes every data-generating process; we call  $\eta$  the moderation signal-to-noise. The endpoints have slope-block risks  $R_{\text{CATE}} = pv$  and  $R_{\text{ATE}} = G = \eta pv$ , crossing at  $\eta = 1$ .

**6.2. The linear oracle and the tuning window.** The natural reference point for any data-driven rule is the best factor an oracle could fix in advance.

**Theorem 2 (Fixed-linear oracle; canonical layer).** In the canonical regime, the fixed-truth risk of the linear factor  $c$  is  $R(c) = c^2 pv + (1 - c)^2 G$ , and: (i) the best fixed linear factor and its risk are

$$c^{\text{lin}} = \frac{G}{G + pv} = \frac{\eta}{1 + \eta}, \quad R(c^{\text{lin}}) = pv \frac{\eta}{1 + \eta}; \quad (16)$$

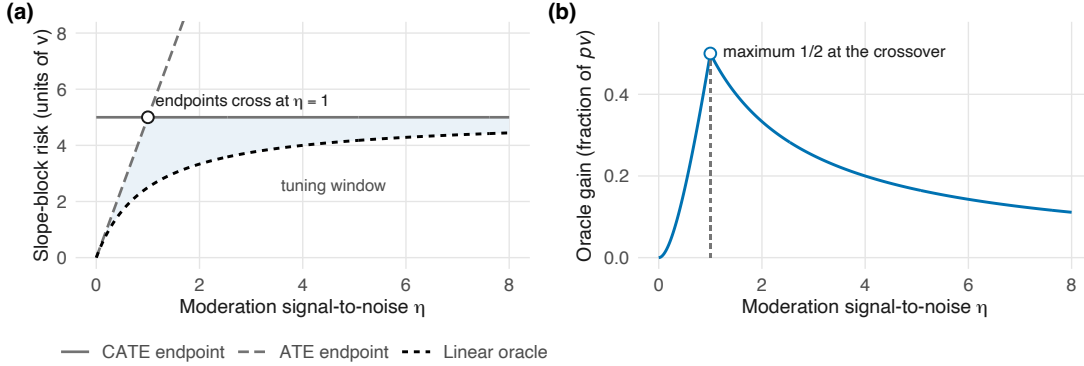


FIGURE 6. The linear oracle and the tuning window (Theorem 2). (a) Endpoint risks and the linear-oracle risk against the moderation signal-to-noise  $\eta$  for  $p = 5$  moderators; the endpoints cross at  $\eta = 1$ . (b) The linear oracle’s gain over the better endpoint as a fraction of the CATE endpoint’s estimation cost; the curve is the same for all  $p$  and  $n$  and attains its maximum of one half at the crossover.

(ii) the linear oracle’s gain over the better endpoint is maximized exactly at the endpoint crossover  $\eta = 1$ , where it equals  $pv/2$ ; and (iii) the gain vanishes as  $\eta \rightarrow 0$  and as  $\eta \rightarrow \infty$ . The proof is given in Appendix E.

Theorem 2 describes where tuning can help, and Figure 6 displays the answer. Tuning is worth the most where the analyst is most uncertain about whether moderation is detectable, because  $\eta = 1$  is the point at which the two endpoints perform equally, and even the linear oracle can save at most half of the moderated predictor’s estimation cost relative to the better endpoint. One caution attends the word oracle:  $c^{\text{lin}}$  is the best *fixed linear scalar* factor, not a lower envelope over all rules. A data-dependent rule can fall below the linear-oracle curve in parts of the signal range, and several rules in Section 7 do. We therefore report every rule’s excess relative to the linear oracle as a comparison, not as a bound.

**6.3. A menu of tuning rules.** Table 3 defines the rules we study. Each is a measurable function of the data alone, each acts on the slope block only, and each amounts to a data-chosen scalar factor on  $\hat{\gamma}_{-1}$ . Three belong to the positive-part family with factor  $(1 - bv/\|\hat{\gamma}_{-1}\|^2)_+$ : the CATE endpoint ( $b = 0$ ), the James–Stein choice ( $b = p - 2$ ), and the empirical-Bayes moment choice ( $b = p$ ).

The hierarchical-Bayes rule uses the posterior mean of the oracle factor under a proper half-normal prior on the conditional signal standard deviation. Its scale is

solved numerically so that the resulting *marginal* prior on  $\gamma_{-1}$  places even odds on the moderation signal exceeding the endpoint crossover,  $\mathbb{P}(\eta > 1) = 1/2$ . We avoid the more common truncated log-uniform hyperprior because its zero-signal risk is determined by the truncation point rather than by the data; in this setting the zero-signal risk moves from  $0.40v$  to  $0.10v$  as the lower cutoff falls from  $10^{-2}$  to  $10^{-8}$ , so any reported premium would be an artifact of the cutoff. Appendix E gives the calibration equation, and Section 7.2 reports the sensitivity of the results to the elicited scale. The pretest rule keeps the CATE endpoint when a size-0.05 chi-square test rejects zero moderation and the ATE endpoint otherwise; we treat it as a stylized comparator for testing-based practice rather than as a measured description of any literature. Five-fold cross-validation operates on the original regression and selects the penalty by held-out outcome error.

**6.4. Dominance over the moderated endpoint.** The first guarantee transfers the classical dominance results to the path.

**Theorem 3 (One-sided dominance and its price; canonical layer).** In the canonical regime with  $p \geq 3$ : (i) for every  $b$  with  $0 < b < 2(p-2)$ , the plain rule with factor  $1 - bv/\|\hat{\gamma}_{-1}\|^2$  satisfies

$$R(\tilde{\gamma}; \gamma_{-1}) = pv - v^2 b \{2(p-2) - b\} \mathbb{E} \left[ \frac{1}{\|\hat{\gamma}_{-1}\|^2} \right] < pv \quad \text{for every } \gamma_{-1}; \quad (17)$$

(ii) the positive-part version has risk strictly smaller than the plain version at every truth, so the James–Stein rule ( $b = p - 2$ ) is strictly better than the CATE endpoint for every truth when  $p \geq 3$ , and the empirical-Bayes rule ( $b = p$ ) when  $p \geq 4$ , while at  $p = 3$  the empirical-Bayes rule violates dominance; (iii) at zero moderation the plain James–Stein rule’s risk is exactly  $2v$  and the positive-part rule’s is strictly smaller, so the premium paid relative to the ATE endpoint at  $\eta = 0$  is at most two coordinates of estimation variance; (iv) for  $p \leq 2$  the plain rules have infinite risk because  $\mathbb{E}[1/\|\hat{\gamma}_{-1}\|^2]$  diverges, and no rule whatsoever is better than the CATE endpoint for every truth, because the unregularized mean is admissible in one and two dimensions under Euclidean loss. Proofs are given in Appendix E.

Theorem 3 has a natural reading in terms of insurance. An analyst who fears the moderated endpoint’s estimation cost can buy protection against it that never backfires, provided there are at least three moderators, and the maximal premium relative to the ATE endpoint is  $2v$  regardless of  $p$ ; the protection therefore becomes more attractive as moderators accumulate. The empirical-Bayes condition  $p \geq 4$  is

TABLE 3. The rule menu. Each rule acts on the slope block  $\hat{\gamma}_{-1}$  only;  $\hat{G} = \|\hat{\gamma}_{-1}\|^2 - pv$  and  $(x)_+ = \max(x, 0)$ . Guarantees are canonical-regime statements. Under target-weighted loss, JS+ requires  $p_{\text{eff}} \geq (p + 2)/2$  and EB/SURE+ requires  $p_{\text{eff}} \geq (p + 4)/2$  (Theorem 4).

| Rule                       | Definition                                                                                                                                    | Guarantee                                                                                                        |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| CATE endpoint              | $\tilde{\gamma} = \hat{\gamma}_{-1}$                                                                                                          | Risk = $pv$ at every truth                                                                                       |
| ATE endpoint               | $\tilde{\gamma} = \mathbf{0}$                                                                                                                 | Risk = $G$ ; unbeatable at $\eta = 0$ (Theorem 5)                                                                |
| James–Stein (JS+)          | Factor $\{1 - (p - 2)v/\ \hat{\gamma}_{-1}\ ^2\}_+$                                                                                           | Strictly better than CATE when $p \geq 3$ ; under target weighting requires $p_{\text{eff}} \geq (p + 2)/2$      |
| EB/SURE                    | Factor $\{1 - pv/\ \hat{\gamma}_{-1}\ ^2\}_+ = \hat{G}_+ / (\hat{G}_+ + pv)$                                                                  | Same guarantee for $p \geq 4$ ; under target weighting requires $p_{\text{eff}} \geq (p + 4)/2$ ; minimizes SURE |
| Hierarchical Bayes         | Factor $\mathbb{E}\{g/(g + v) \mid \hat{\gamma}_{-1}\}$ under a half-normal prior on $\sqrt{g}$ elicited so that $\mathbb{P}(\eta > 1) = 1/2$ | None claimed; best average behavior on our grid; prior-scale sensitivity reported                                |
| Pretest ( $\alpha = .05$ ) | CATE endpoint if $\ \hat{\gamma}_{-1}\ ^2/v > \chi_{p,.95}^2$ , else ATE endpoint                                                             | None; post-selection risk bump (Leeb & Pötscher, 2005); stylized comparator                                      |
| Five-fold CV               | Penalty by held-out outcome error on the raw regression                                                                                       | None; practice-standard comparator (finite-design track)                                                         |
| Linear oracle (reference)  | Factor $\eta/(1 + \eta)$ , uses the truth                                                                                                     | Best fixed linear factor (Theorem 2); not a lower envelope over all rules; not implementable                     |

sharp: deterministic quadrature exhibits the violation at  $p = 3$  (risk  $3.19v$  against the endpoint’s  $3v$  at  $\eta = 4$ ) and confirms strict dominance at  $p = 4$ , where the zero-moderation risk is  $4 + \mathbb{E}[(Z - 8)\mathbf{1}\{Z \leq 4\}] = 0.54$  in units of  $v$ , with  $Z \sim \chi_4^2$  (Appendix F). Clause (iv) matters for applied work because two-moderator analyses are common; with  $p \leq 2$  the endpoint choice remains genuinely binary.

**6.5. Target-weighted loss and the effective dimension.** Theorem 3 is proved under matched target moments. When predictions are needed for a mismatched target, the slope loss becomes weighted,  $\|\tilde{\gamma}_{-1} - \gamma_{-1}\|_{\mathbf{W}}^2 = (\tilde{\gamma}_{-1} - \gamma_{-1})^\top \mathbf{W} (\tilde{\gamma}_{-1} - \gamma_{-1})$ , where  $\mathbf{W} \succ \mathbf{0}$  is the slope block of the whitened target second moment, the same matrix whose trace is the slope part of the relative leverage  $L_{T|S}$ . Euclidean

dominance does not transfer automatically, and the next theorem gives the transfer its exact condition.

**Theorem 4 (Bock–Berger specialization for target-weighted loss; canonical layer).** In the canonical regime with  $p \geq 3$  and slope loss weighted by  $\mathbf{W} \succ \mathbf{0}$ , let  $p_{\text{eff}} = \text{tr}(\mathbf{W})/\lambda_{\max}(\mathbf{W})$  denote the effective dimension of the target (Bock, 1975). For the plain rule with factor  $1 - bv/\|\hat{\gamma}_{-1}\|^2$ : (i) the weighted risk satisfies the identity

$$R_{\mathbf{W}}(\tilde{\gamma}; \gamma_{-1}) = v \text{tr}(\mathbf{W}) + v^2 \mathbb{E} \left[ -\frac{2b \text{tr}(\mathbf{W})}{\|\hat{\gamma}_{-1}\|^2} + \frac{(4b + b^2) \hat{\gamma}_{-1}^\top \mathbf{W} \hat{\gamma}_{-1}}{\|\hat{\gamma}_{-1}\|^4} \right]; \quad (18)$$

(ii) if  $0 < b < 2(p_{\text{eff}} - 2)$ , the plain rule is strictly better than the CATE endpoint at every truth; at the boundary  $b = 2(p_{\text{eff}} - 2) > 0$  it is weakly better. In both cases the positive-part version strictly improves the plain rule and is therefore strictly better than the endpoint; (iii) at zero moderation the  $b = p - 2$  rule’s weighted risk is exactly  $2v \text{tr}(\mathbf{W})/p$ , two effective coordinates of target-weighted estimation variance; (iv) if  $p_{\text{eff}} \leq 2$ , the window in (ii) is empty, and no fixed  $b > 0$  in the Euclidean-factor family is weakly better than the CATE endpoint for all truths, because along the top eigendirection of  $\mathbf{W}$  the excess risk is asymptotically  $v^2 b \lambda_{\max}(\mathbf{W}) \{b + 4 - 2p_{\text{eff}}\}/G > 0$  as the signal grows; (v) consequently, the actual menu constants have stronger conditions than window existence: JS+ ( $b = p - 2$ ) is protected when  $p_{\text{eff}} \geq (p + 2)/2$ , whereas EB/SURE+ ( $b = p$ ) is protected when  $p_{\text{eff}} \geq (p + 4)/2$ . The weaker condition  $p_{\text{eff}} > 2$  says only that some positive  $b$  is available. Proofs are given in Appendix E; the result is a target-design specialization of the general-covariance and general-quadratic-loss theory of Bock (1975) and Berger (1976).

Theorem 4 connects the two halves of the article. The matrix  $\mathbf{W}$  is the whitened target moment whose trace prices the estimation component in Proposition 3; the Bock–Berger theory then turns its shape into rule-specific protection conditions. At  $p = 6$ , for example, the mildly concentrated target in Figure 8(b) has  $p_{\text{eff}} = 2.67$ , so a small protected  $b$  exists, but neither the JS constant  $b = 4$  nor the EB/SURE constant  $b = 6$  is protected. The premium formula in (iii) remains exact, but it is not a substitute for checking (v). Note that the empty-window result concerns only rules whose scalar factor uses the Euclidean norm; a rule adapted directly to the known  $\mathbf{W}$ -geometry is outside this article’s menu and is not ruled out. For design, the practical step is to compute both  $\text{tr}(\mathbf{W})$  and  $p_{\text{eff}}$  and then check the condition for the rule actually planned.

**6.6. A bound on all tuning rules.** The guarantees above are one-sided, and this is not an artifact of the menu.

**Theorem 5 (No free lunch on the path; canonical layer).** In the canonical regime with  $v > 0$ , let  $\tilde{\gamma}(\cdot)$  be any measurable estimator of  $\gamma_{-1}$ . If its risk at  $\gamma_{-1} = \mathbf{0}$  equals the ATE endpoint's risk there, which is zero, then  $\tilde{\gamma} = \mathbf{0}$  Lebesgue-almost everywhere, so the rule is the ATE endpoint up to almost-everywhere equivalence and its risk is  $G$  at every truth. Consequently no measurable rule is weakly better than both endpoints for all truths, and every rule other than the ATE endpoint pays a strictly positive premium at zero moderation. The proof, an equivalence-of-Gaussian-measures argument, is given in Appendix E.

Theorem 5 applies to every possible answer to the tuning question, including answers we did not study, because the proof uses nothing about linearity or rotation invariance. The oracle frontier of Proposition 2 cannot be attained uniformly by any data-driven procedure, so the design question is not which rule wins everywhere but which profile of premiums fits the setting: what a rule pays at zero moderation against the excess it suffers relative to the linear oracle elsewhere. We emphasize that the theorem is an elementary calibration result rather than a deep impossibility theorem; its role is to keep the premium language honest.

**6.7. Unbiased risk estimation.** It remains to describe how the data can adjudicate between the endpoints without an appeal to the prior.

**Proposition 5 (SURE on the path; canonical layer).** In the canonical regime with  $v$  known: (i) for every prespecified factor  $c$ , the statistic

$$\text{SURE}(c) = (1 - c)^2 \|\hat{\gamma}_{-1}\|^2 + (2c - 1) p v \quad (19)$$

satisfies  $\mathbb{E}\{\text{SURE}(c)\} = R(c)$  at every truth, so the fixed-truth risk that the prior-trusting design criterion cannot see is estimable without bias from the data alone; (ii) the minimizer of (19) over  $c \in [0, 1]$  is the empirical-Bayes rule of Table 3, an identity already noted in the homoscedastic hierarchical setting by Xie et al. (2012), so continuous SURE tuning and moment empirical Bayes are the same rule on this path; (iii) writing  $T = \|\hat{\gamma}_{-1}\|^2$  and  $\hat{c} = (1 - pv/T)_+$ , the selection optimism is exact:

$$\mathbb{E}\{\min_c \text{SURE}(c)\} = R(\hat{c} \hat{\gamma}_{-1}) - 4pv^2 \mathbb{E}\{T^{-1} \mathbf{1}(T > pv)\}.$$



The correction is strictly positive, so the minimized value is not an unbiased estimate of the selected rule’s risk; reporting that risk requires a selection-aware method such as cross-fitting or an independent validation split. Proofs are given in Appendix E.

Proposition 5 closes the circle left open in Section 4.2. The obstruction there was that the EPPV under the path’s own prior is monotone in the penalty; SURE breaks the circularity by estimating the fixed-truth risk instead of averaging over the prior, and it requires no new machinery, because its minimizer is a rule already on the menu. Part (iii) is the boundary we attach so that the resolution is not overstated.

## 7. Simulation Study

**7.1. Design.** The study has two tracks. The canonical track evaluates the full menu in the regime of Lemma 2 at dimensions  $p \in \{2, 5, 10\}$  over a thirteen-point signal grid  $\eta \in [0, 8]$ . Its primary evidence is deterministic: every spherical-factor rule’s risk, including the hierarchical rule’s, is computed by adaptive noncentral chi-square quadrature with checked error diagnostics, and a full Monte Carlo replication at 200,000 draws per cell runs alongside as an agreement check, with worst relative deviation 1.3% (Appendix F).

The finite-design track generates full experiments from the interaction regression (10) with balanced arms  $n \in \{20, 50, 100, 200\}$  at  $p = 5$ , estimates the residual scale, whitens with the estimated slope covariance, applies the same rules, and evaluates the Euclidean slope loss in original coordinates. The linear comparator in this track is the data-dependent plug-in factor  $c_E^{\text{lin}} = \|\gamma_{-1}\|^2 / (\|\gamma_{-1}\|^2 + \text{tr } \mathbf{V}_s)$ , computed and evaluated in the same Euclidean metric, where  $\mathbf{V}_s$  uses the realized design and the outcome-estimated residual scale. The track adds five-fold cross-validation and a heteroscedastic variant with arm scales (1, 2) fitted under the common-scale convention. Every reported number is generated by the verification and simulation suite with fixed seeds, and the suite labels each block by its evidence type (analytic proof, deterministic quadrature, independent Monte Carlo, regression check, or descriptive report), so that the kind of evidence supporting each claim is explicit.

**7.2. Canonical results.** Figure 7 displays the menu at  $p = 5$ , and Table 4 reports the premium at zero moderation, the maximum excess over the linear oracle across the signal grid (a grid maximum, not a continuum supremum), and the maximum excess over the better endpoint, for every rule and dimension.

The guaranteed rules behave as the theory predicts. The James–Stein and empirical-Bayes rules run below the CATE endpoint at every grid signal, in accordance with Theorem 3 for  $p \geq 3$  and  $p \geq 4$ . At the crossover  $\eta = 1$ , where the linear-oracle risk is  $2.50v$ , they attain  $3.50v$  and  $3.40v$  against the endpoints’  $5.0v$ . Their premiums at zero moderation are  $1.40v$  and  $0.57v$ .

The elicited hierarchical rule is the strongest all-around performer on this grid, although it carries no dominance guarantee. Its zero-moderation premium is  $0.71v$ , its risk at the crossover is  $2.95v$ , the best on the menu, and its maximum grid excess over the linear oracle is  $0.86v$ , smaller than the empirical-Bayes rule’s  $1.01v$  and the James–Stein rule’s  $1.40v$ . Unlike its guaranteed competitors it can in principle exceed the CATE endpoint, but on this grid it never does at  $p = 5$ . Halving or doubling the elicited prior scale moves the zero-signal premium from  $0.39v$  through  $0.71v$  to  $0.90v$ , a max/min ratio of 2.32; this sensitivity is material, and we report it rather than describe the rule as stable. The elicited construction nevertheless has an advantage over a cutoff-driven hyperprior: the rule is tied to an explicit marginal probability statement, and the sensitivity curve shows what the elicitation choice costs.

The pretest rule performs worst in the region where tuning has the most to offer. Its risk exceeds the better endpoint through the middle of the window, by a maximum of  $1.90v$  at  $\eta = 2$  (risk  $6.90v$  against  $5.0v$ ); it exceeds both endpoints at the crossover itself ( $5.69v$ ); and its maximum grid excess over the linear oracle ( $3.68v$  at  $p = 5$ ,  $5.93v$  at  $p = 10$ ) is three to six times that of the tuned rules. The mechanism is the familiar post-selection one (Leeb & Pötscher, 2005): near the detection boundary the test keeps choosing the wrong endpoint with non-trivial probability. An analyst who plans to test for moderation and then choose therefore adopts the procedure with the worst regret profile in the region where the choice is genuinely uncertain, and moderate undetected moderation is arguably the typical situation in applied experiments (Weiss et al., 2017).

Dimension moves the menu as Theorem 3 predicts; Appendix F displays  $p = 2$  and  $p = 10$ . At  $p = 2$  no rule is uniformly safe and the James–Stein rule degenerates to the CATE endpoint; the empirical-Bayes and hierarchical rules still perform well on the grid, but their good behavior is unprotected. At  $p = 10$  the guarantees bind, and every tuned rule tracks the oracle over a growing share of the grid.

**7.3. Finite designs, estimated scale, and cross-validation.** Figure 9 compares the canonical curves with the homoscedastic finite-design track. Across all twelve JS+, EB/SURE, and hierarchical-Bayes cells at each arm size ( $\eta \in \{0, .25, 1, 4\}$ ),

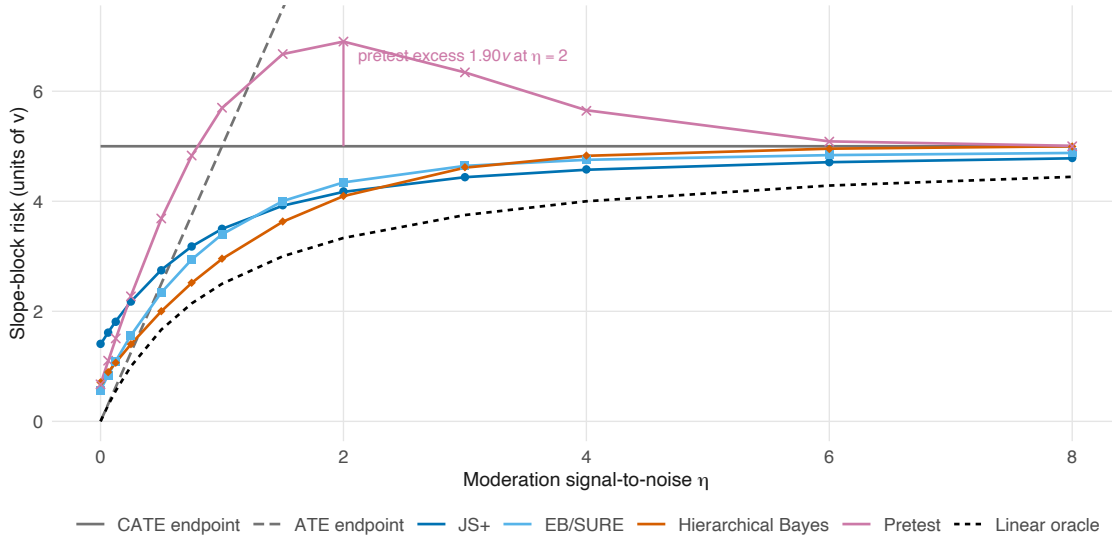


FIGURE 7. Slope-block risk of the rule menu in the canonical regime at  $p = 5$ , in units of the per-coordinate estimation variance  $v$ . Lines are deterministic quadrature values; points are the 200,000-replication Monte Carlo agreement track; grey lines are the endpoints; the dashed line is the linear oracle of Theorem 2. The pretest rule exceeds the better endpoint through the tuning window, with maximal excess  $1.90v$  at  $\eta = 2$ ; the tuned rules respect the guarantees of Theorem 3.

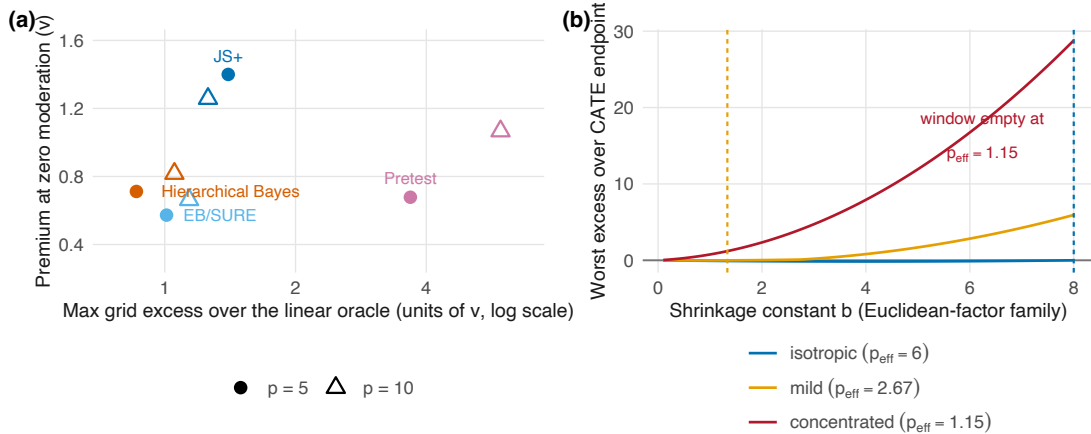


FIGURE 8. Premiums and the target-weighted window. (a) Zero-signal premium against maximum grid excess over the linear oracle at  $p = 5$  and  $p = 10$ ; every implementable rule pays a positive premium. (b) Worst excess over the CATE endpoint across a signal probe set for three target shapes at  $p = 6$ ; dotted verticals mark the window boundary  $2(p_{\text{eff}} - 2)$ , and the Euclidean-factor window is empty at  $p_{\text{eff}} = 1.15$ .

TABLE 4. Insurance coordinates of the rule menu (canonical layer, units of  $v$ , deterministic quadrature). The premium is the risk at  $\eta = 0$ , where the ATE endpoint is unbeatable. The excess columns are maxima over the thirteen-point signal grid: relative to the linear oracle of Theorem 2 and relative to the better endpoint. At  $p = 2$  the James–Stein rule coincides with the CATE endpoint (empty Stein window).

| $p$ | Rule          | Premium<br>at $\eta = 0$ | Max grid excess<br>vs oracle | Max excess vs<br>better endpoint |
|-----|---------------|--------------------------|------------------------------|----------------------------------|
| 2   | CATE endpoint | 2.00                     | 2.00                         | 2.00                             |
| 2   | ATE endpoint  | 0                        | 14.22                        | 14.00                            |
| 2   | JS+ (= CATE)  | 2.00                     | 2.00                         | 2.00                             |
| 2   | EB/SURE       | 0.44                     | 0.81                         | 0.44                             |
| 2   | Hierarchical  | 0.53                     | 0.64                         | 0.53                             |
| 2   | Pretest       | 0.40                     | 2.25                         | 1.75                             |
| 5   | CATE endpoint | 5.00                     | 5.00                         | 5.00                             |
| 5   | ATE endpoint  | 0                        | 35.56                        | 35.00                            |
| 5   | JS+           | 1.40                     | 1.40                         | 1.40                             |
| 5   | EB/SURE       | 0.57                     | 1.01                         | 0.57                             |
| 5   | Hierarchical  | 0.71                     | 0.86                         | 0.71                             |
| 5   | Pretest       | 0.68                     | 3.68                         | 1.90                             |
| 10  | CATE endpoint | 10.00                    | 10.00                        | 10.00                            |
| 10  | ATE endpoint  | 0                        | 71.11                        | 70.00                            |
| 10  | JS+           | 1.26                     | 1.26                         | 1.26                             |
| 10  | EB/SURE       | 0.66                     | 1.14                         | 0.66                             |
| 10  | Hierarchical  | 0.82                     | 1.05                         | 0.82                             |
| 10  | Pretest       | 1.07                     | 5.93                         | 2.47                             |

finite-to-canonical relative deviations range from 23.2% to 84.6% at  $n = 20$ , from 7.3% to 17.8% at  $n = 50$ , from  $-0.4\%$  to  $8.5\%$  at  $n = 100$ , and from  $-0.8\%$  to  $5.0\%$  at  $n = 200$ . Every tuned-rule cell is therefore within  $8.5\%$  of its canonical value from  $n = 100$  onward, although describing the agreement as “a few percent from  $n = 100$ ” would be too strong. The ordering of the tuned rules matches the canonical ordering in every cell from  $n = 100$  onward; at  $n = 20$  and  $50$  it differs at  $\eta = 4$ . Appendix F reports the complete cell table, the heteroscedastic panel, and the five-fold cross-validation results.

## 8. Design Implications

The results above have three practical implications.

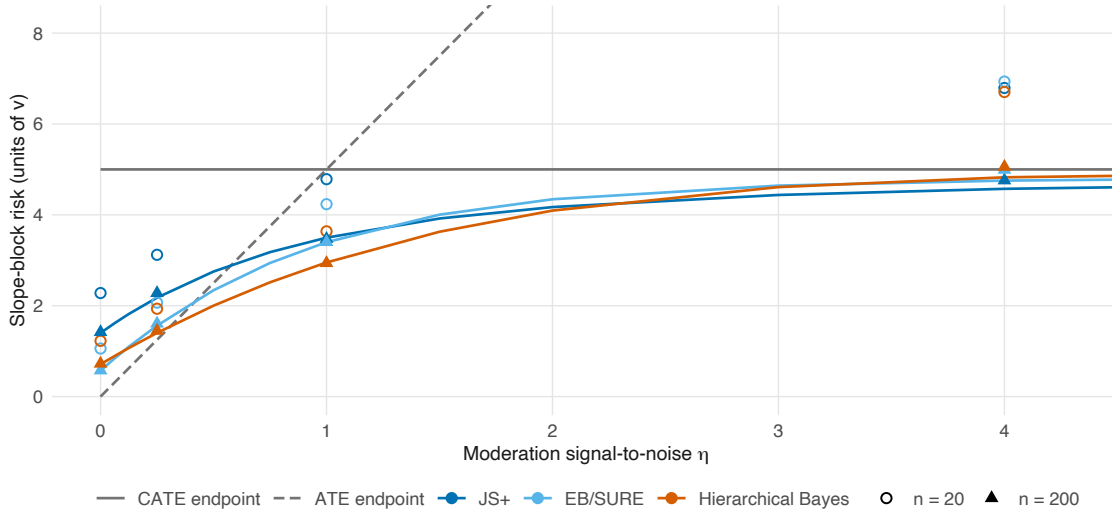


FIGURE 9. Homoscedastic finite-design validation at  $p = 5$  (points; 2,000 replications per cell) against canonical quadrature curves (lines). Residual scale is estimated in every replication. The full  $n$ -by- $\eta$  comparison is reported in Appendix F.

First, the endpoint gamble can be retired when the planned rule satisfies its dimension condition. A designer who anticipates tuning does not need to decide at the design stage whether the analysis will be average-effect or moderated; the data will place the analysis on the path, and Theorem 3 bounds the cost at zero moderation. For a mismatched target, however,  $p_{\text{eff}} > 2$  only makes the protected window nonempty: the JS+ rule requires  $p_{\text{eff}} \geq (p+2)/2$ , and EB/SURE+ requires  $p_{\text{eff}} \geq (p+4)/2$ . When the relevant condition holds, Theorem 4 gives the zero-signal premium  $2v \text{tr}(\mathbf{W})/p$ . The design-stage question is therefore both a premium question and a rule-specific window check.

Second, anticipation can be made quantitative with a preposterior calculation, and the calculation returns a differentiated verdict. Figure 10(b) reports the expected regret of each rule relative to the linear oracle under three anticipation scenarios encoded as distributions over  $\eta$ : a skeptic (uniform over the grid points with  $\eta \leq 0.5$ ), an agnostic ( $\eta \leq 4$ ), and an optimist ( $\eta \geq 1$ ). The marginally calibrated hierarchical rule has expected regret  $0.51v$ ,  $0.59v$ , and  $0.68v$  in the three scenarios and remains the best tuned rule in all of them. For the agnostic and the optimist, every tuned rule beats both endpoints, so a designer who anticipates a real possibility of moderation no longer needs to gamble on an endpoint. For the skeptic, the ATE endpoint remains best at  $0.23v$ , and even the cheapest implementable protection, here the hierarchical

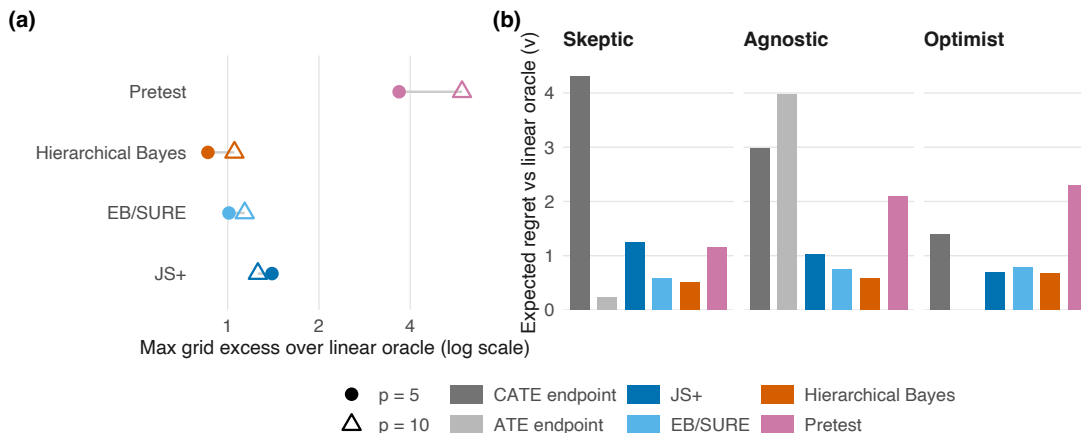


FIGURE 10. Regret profiles of the rule menu (canonical regime,  $p = 5$ ). (a) Maximum grid excess of each rule relative to the linear oracle, on a logarithmic scale. (b) Preposterior expected regret under three anticipation scenarios over the moderation signal. The tuned rules beat both endpoints for the agnostic and the optimist; for the skeptic the ATE endpoint remains best, and protection carries a visible price.

rule at  $0.51v$ , costs about  $0.28v$  relative to that commitment. Tuning is thus close to free for agnostics and optimists, and a priced insurance policy for skeptics.

Third, moderator-count planning acquires a new term at the zero-moderation coordinate. In the untuned calculus each moderator costs one unit of estimation variance in the moderated analysis; with a tuned analysis the total cost of carrying candidate moderators that turn out to be inert is capped at  $2v$  by Theorem 3, not  $v$  per moderator, and the mismatched-target version of the cap is  $2v \text{tr}(\mathbf{W})/p$ . Away from zero moderation the theorems are silent and regret can still grow with  $p$  (Table 4), so the cap is a statement about dead weight rather than a blanket license to add moderators, and the finite-design track shows the realized cost of small  $n$  with many moderators. A caution runs the other way at  $p \leq 2$  or  $p_{\text{eff}} \leq 2$ : tuning offers no protection there, and designs built around a single high-stakes moderator should plan for the pretest problem rather than assume that shrinkage will absorb it.

## 9. Discussion

We have developed the Bayesian side of the unit-specific prediction design program in one arc, from criterion to bridge to implementable tuning. The first result is the bridge itself. Under the conditional frame with diffuse priors, the EPPV recovers the Tipton–Mamakos benchmark exactly to first order, the finite-sample gap is the

closed-form correction of Theorem 1, and Corollary 1 extends the exact arithmetic to arbitrary Gaussian sources and arbitrary targets, so cross-population planning numbers that previously lived at first order can now be computed exactly. The correction has a dual frequentist reading, and we present it as an audit of the population-moment approximation rather than as a Bayesian advantage.

The second set of results concerns the design levers, each of which comes with a boundary that planning should respect. Prior precision provably reduces the estimation component, but a half- $\sigma$  offset in the prior mean erases the gain by  $n_0 \approx 30$  in our default setting. The shrinkage continuum reframes the model choice as a path, but half of the interior choices on our grid are worse than the better endpoint. Relative leverage can fall below its representative benchmark while total risk still rises through source noise, a decoupling that the domain-indexed residual model of Assumption 1 makes explicit. The unidentified correlation leaves fixed-floor design rankings alone, but it decides feasibility, and it moves interval coverage asymmetrically when misassumed.

The third set of results concerns tuning. One-sided protection is available and inexpensive under its conditions: with three or more moderators, positive-part shrinkage protects against the moderated endpoint’s estimation cost at a premium of at most  $2v$  at zero moderation. For mismatched targets, the Bock–Berger specialization uses the whitened target moment both to price that premium and to check the chosen rule’s effective-dimension condition. Two-sided protection is impossible, so every rule pays a positive premium somewhere, and the choice among rules is a choice among premium profiles. Unbiased risk estimation is the appropriate adjudication layer, because SURE estimates the fixed-truth risk that the prior-trusting criterion cannot see, with an exact positive correction for the optimism of its minimized value. The practice-shaped comparator, pretesting for moderation, performs worst in the window where the moderation question is genuinely uncertain.

Throughout, the risk layers matter as much as the results, and in our experience conflating them is the most common route to overclaiming in this area. The EPPV is monotone in the prior it trusts and cannot referee the endpoints; the fixed-truth layer can, and SURE estimates it; the canonical guarantees are exact in a declared regime whose distance to realized designs the finite-design track prices cell by cell. Appendix F collects the claim boundaries and evidence types in a form intended to be reusable.

The limitations trace the assumptions. All results are conditional on working residual parameters; integrating over them couples the risk to the non-identified  $\rho_T$  and is left to future work. The working model is linear and Gaussian, chosen for closed-form design arithmetic; flexible learners can replace it inside the same criterion by simulation, and how much of the tuning guarantees survives outside the canonical regime is an open question, although the SURE logic and Theorem 5 both extend in spirit. The transport condition (A5) rules out coefficient drift, which real source-target pairs need not obey. The rotation-invariant rules respond only to the total moderation signal, so sparse alternatives trade the uniform guarantees for adaptivity we did not evaluate. The pretest comparator uses a single conventional size. Cluster designs satisfy (A2) at the site level, and multilevel extensions remain open. Finally, sample-side design, the choice of source distribution and sites for a stated target, is developed in the companion article (Lee, 2026), which takes the leverage and effective-dimension objects of this article as its starting point.

For practice, we suggest the following. Plan with the exact formulas at small arm sizes, because the first-order benchmark understates risk precisely where samples are expensive. State prior evidence as an effective sample size with an offset scenario attached. Report feasibility over a stated range of  $\rho_T$ . Tune rather than test, but only after checking the planned rule: JS+ requires  $p \geq 3$  and, under target weighting,  $p_{\text{eff}} \geq (p + 2)/2$ , whereas EB/SURE+ requires  $p \geq 4$  in the matched case and  $p_{\text{eff}} \geq (p + 4)/2$  in general. Use the elicited hierarchical rule when average performance across an anticipated signal range is the objective and a worst-case guarantee is not required. If only  $p_{\text{eff}} > 2$  is known, a protected scalar  $b$  exists but the standard menu constants need not qualify, and with one or two moderators, or a target whose effective dimension is at most two, the endpoint decision remains outside this Euclidean-factor guarantee. In reports, state the realized path position alongside the predictions, because a tuned analysis that landed near an endpoint conveys what a pretest would have asserted with false confidence.

**Reproducibility.** All numerical claims in this article, including every figure, table, and in-text value, are generated by a documented verification and simulation suite. The canonical risk computations are deterministic noncentral chi-square quadrature; Monte Carlo is used for the finite-design, coverage, and agreement tracks, with fixed seeds and no Markov chain Monte Carlo. Every verification block carries an evidence-type label (analytic proof, deterministic quadrature, independent Monte Carlo, regression or transcription check, or descriptive report), so the verification map



in Appendix F states not only that each formal claim was checked but what kind of evidence the check provides. The suite (R scripts `00_utils.R`, `01_verification.R`, `02_simulation.R`, and `03_figures.R`, with a README) and the verification report from the shipped run accompany this article as arXiv ancillary files.

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# Online Supplement

## Designing Experiments to Predict Unit-Specific Treatment Effects: A Bayesian Bridge with Implementable Tuning

JoonHo Lee

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## Appendix A. Setup and Proofs of Lemma 1, Theorem 1, and Corollary 1

**A.1. Setup and notation.** Units carry potential outcomes  $(Y_i(0), Y_i(1))$ , moderators  $\mathbf{x}_i \in \mathbb{R}^p$ , and the augmented vector  $\tilde{\mathbf{x}}_i = (1, \mathbf{x}_i^\top)^\top$  with  $q = p + 1$ . The source working model is

$$Y_i(a) = \tilde{\mathbf{x}}_i^\top \boldsymbol{\beta}_a + \varepsilon_i(a), \quad \text{Var}\{\varepsilon_i(a) \mid \mathbf{x}_i\} = \sigma_{a,S}^2, \quad (\text{A.1})$$

with the residual pair jointly Gaussian given  $\mathbf{x}_i$ . In the target population the same coefficients govern the conditional means (main-text Assumption 1, part A5), and the target residual pair has variances  $\sigma_{a,T}^2$  and correlation  $\rho_T$ , giving the floor  $\tau_{T|x}^2(\rho_T) = \sigma_{1,T}^2 + \sigma_{0,T}^2 - 2\rho_T\sigma_{1,T}\sigma_{0,T}$ . Write  $\mathbf{S}_a = \tilde{\mathbf{X}}_a^\top \tilde{\mathbf{X}}_a$  for the augmented cross-product in arm  $a$  and

$$\pi(\boldsymbol{\beta}_a \mid \Sigma_\varepsilon) \propto \exp\left\{-\frac{1}{2\sigma_{a,S}^2}(\boldsymbol{\beta}_a - \mathbf{m}_a)^\top \mathbf{K}_a(\boldsymbol{\beta}_a - \mathbf{m}_a)\right\}, \quad \mathbf{K}_a \succeq \mathbf{0}, \quad (\text{A.2})$$

independent across arms. The prior is proper if and only if  $\mathbf{K}_a \succ \mathbf{0}$ ;  $\mathbf{K}_a = \mathbf{0}$  is the flat prior. Throughout this appendix the propriety condition (A8) is in force:  $\mathbf{S}_a + \mathbf{K}_a \succ \mathbf{0}$  almost surely. Under a flat prior this requires  $\mathbf{S}_a \succ \mathbf{0}$ , which holds almost surely when  $n_a \geq q$  and the covariate distribution assigns zero mass to every proper linear subspace; under a partially flat prior with  $\mathbf{K}_a$  singular, the design must span the null space of  $\mathbf{K}_a$ , because on that null space only the likelihood curvature is available. When the propriety condition fails, the posterior is improper and no statement below is defined; the case  $n_a < q$  under a flat prior is the practically relevant failure.

**A.2. Proof of Lemma 1. Lemma 1 (restated; conditional posterior-predictive variance).** *Under Assumption 1, for a future target unit with augmented*

covariates  $\tilde{\mathbf{x}}$ ,

$$\text{Var}(\delta \mid D, \tilde{\mathbf{x}}, \Sigma_\varepsilon) = \sigma_{1,S}^2 \tilde{\mathbf{x}}^\top (\mathbf{S}_1 + \mathbf{K}_1)^{-1} \tilde{\mathbf{x}} + \sigma_{0,S}^2 \tilde{\mathbf{x}}^\top (\mathbf{S}_0 + \mathbf{K}_0)^{-1} \tilde{\mathbf{x}} + \tau_{T|x}^2(\rho_T), \quad (\text{A.3})$$

and the right side does not depend on the observed outcomes.

*Proof.* Randomization makes treatment independent of the potential outcomes given the covariates, so the observed-data likelihood factors over arms (equation (D.1) of Appendix D), and with arm-independent priors the joint posterior of  $(\beta_0, \beta_1)$  is the product of the arm-wise posteriors. Completing the square in the exponent of likelihood times prior for arm  $a$  gives the conjugate Gaussian update

$$\beta_a \mid D, \Sigma_\varepsilon \sim \mathbf{N}_q \left( (\mathbf{S}_a + \mathbf{K}_a)^{-1} (\tilde{\mathbf{X}}_a^\top \mathbf{y}_a + \mathbf{K}_a \mathbf{m}_a), \sigma_{a,S}^2 (\mathbf{S}_a + \mathbf{K}_a)^{-1} \right), \quad (\text{A.4})$$

which is well defined by (A8). For the future unit,  $\delta = \tilde{\mathbf{x}}^\top \beta_1 - \tilde{\mathbf{x}}^\top \beta_0 + \nu$ , where  $\nu$  is the unit's residual effect contrast. The three terms are mutually independent given  $(D, \tilde{\mathbf{x}}, \Sigma_\varepsilon)$ : the two coefficient terms because the posterior factors over arms, and  $\nu$  because the future unit is not in the experiment and (A5) fixes its conditional law with variance  $\tau_{T|x}^2(\rho_T)$ . Variances therefore add,

$$\text{Var}(\delta \mid D, \tilde{\mathbf{x}}, \Sigma_\varepsilon) = \tilde{\mathbf{x}}^\top \text{Var}(\beta_1 \mid D) \tilde{\mathbf{x}} + \tilde{\mathbf{x}}^\top \text{Var}(\beta_0 \mid D) \tilde{\mathbf{x}} + \tau_{T|x}^2(\rho_T),$$

and substituting the posterior covariances from (A.4) gives (A.3). The posterior covariance in (A.4) depends on the data only through  $\mathbf{S}_a$ , which is a function of the design points alone, so the right side of (A.3) is free of the outcomes.  $\square$

**Remark A.1 (Coupled priors).** If the arms are given a joint prior with cross-arm covariance, the posterior acquires cross-arm blocks  $\mathbf{C}_{aa'} = \text{Cov}(\beta_a, \beta_{a'} \mid D)$  and the coefficient part of the predictive variance becomes  $\tilde{\mathbf{x}}^\top (\mathbf{C}_{11} + \mathbf{C}_{00} - \mathbf{C}_{10} - \mathbf{C}_{01}) \tilde{\mathbf{x}}$ . The sign of the correction relative to (A.3) depends on the prior coupling, and we do not pursue it.

**A.3. Proof of Theorem 1. Theorem 1 (restated; conditional bridge).** Under Assumption 1 with  $T = S$ , diffuse priors, whitened Gaussian design rows, and  $n_a > p + 2$ ,

$$R_c(d \mid \Sigma_\varepsilon) = \sigma_0^2 \ell_p(n_0) + \sigma_1^2 \ell_p(n_1) + \tau_{S|x}^2(\rho), \quad \ell_p(n) = \frac{1}{n} + \frac{p(1 + 1/n)}{n - p - 2},$$

and  $\ell_p(n) - q/n = p(p + 3)/\{n(n - p - 2)\}$ , which is strictly positive for  $p \geq 1$  and zero for  $p = 0$ .



*Proof.* We first record the diffuse limit. For  $n_a \geq q$  the matrix  $\mathbf{S}_a$  is almost surely positive definite, and along any monotone path  $\mathbf{K}_a \downarrow \mathbf{0}$  in the Loewner order the inverses  $(\mathbf{S}_a + \mathbf{K}_a)^{-1}$  increase monotonically to  $\mathbf{S}_a^{-1}$ , so monotone convergence exchanges the limit with the design expectation; the limit is finite for  $n_a > p + 2$  by the calculation below. For an arbitrary positive semidefinite sequence  $\mathbf{K}_a \rightarrow \mathbf{0}$ , pointwise convergence of the inverses together with the almost-sure bound  $(\mathbf{S}_a + \mathbf{K}_a)^{-1} \preceq \mathbf{S}_a^{-1}$  and integrability of  $\text{tr}(\mathbf{M}_T \mathbf{S}_a^{-1})$  gives the same conclusion by dominated convergence.

Fix one arm and drop the subscript. Write the design rows as  $(1, \mathbf{x}_i^\top)$  with  $\mathbf{x}_i \stackrel{\text{i.i.d.}}{\sim} \mathbf{N}_p(\mathbf{0}, \mathbf{I}_p)$  after whitening,  $\bar{\mathbf{x}} = n^{-1} \sum_i \mathbf{x}_i$ , and

$$\mathbf{G} = \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top.$$

Block inversion of

$$\mathbf{S} = \begin{pmatrix} n & n\bar{\mathbf{x}}^\top \\ n\bar{\mathbf{x}} & \mathbf{X}^\top \mathbf{X} \end{pmatrix}, \quad \mathbf{X}^\top \mathbf{X} = \mathbf{G} + n\bar{\mathbf{x}}\bar{\mathbf{x}}^\top,$$

through the Schur complement of the  $(1, 1)$  entry gives

$$(\mathbf{S}^{-1})_{11} = \frac{1}{n} + \bar{\mathbf{x}}^\top \mathbf{G}^{-1} \bar{\mathbf{x}}, \quad (\mathbf{S}^{-1})_{22} = \mathbf{G}^{-1}, \quad (\mathbf{S}^{-1})_{21} = -\mathbf{G}^{-1} \bar{\mathbf{x}}, \quad (\text{A.5})$$

so that  $\text{tr}(\mathbf{S}^{-1}) = 1/n + \bar{\mathbf{x}}^\top \mathbf{G}^{-1} \bar{\mathbf{x}} + \text{tr}(\mathbf{G}^{-1})$ . For Gaussian rows,  $\mathbf{G} \sim \mathbf{W}_p(n-1, \mathbf{I}_p)$ ,  $\mathbf{G}$  is independent of  $\bar{\mathbf{x}} \sim \mathbf{N}_p(\mathbf{0}, \mathbf{I}_p/n)$  (Muirhead, 1982, Theorem 3.1.2), and the inverse-Wishart mean is  $\mathbb{E}(\mathbf{G}^{-1}) = \mathbf{I}_p/(n-p-2)$  for  $n > p+2$  (Muirhead, 1982, Theorem 3.2.12). Independence gives

$$\mathbb{E}(\bar{\mathbf{x}}^\top \mathbf{G}^{-1} \bar{\mathbf{x}}) = \text{tr}\{\mathbb{E}(\mathbf{G}^{-1}) \mathbb{E}(\bar{\mathbf{x}}\bar{\mathbf{x}}^\top)\} = \frac{p}{n(n-p-2)},$$

so

$$\mathbb{E} \text{tr}(\mathbf{S}^{-1}) = \frac{1}{n} + \frac{p}{n(n-p-2)} + \frac{p}{n-p-2} = \frac{1}{n} + \frac{p(1+1/n)}{n-p-2} = \ell_p(n).$$

Substituting into the trace form of the risk with  $\mathbf{M}_T = \mathbf{I}_q$  and adding the floor gives the exact display. The correction is

$$\ell_p(n) - \frac{p+1}{n} = \frac{p(1+1/n)}{n-p-2} - \frac{p}{n} = \frac{p(n+1) - p(n-p-2)}{n(n-p-2)} = \frac{p(p+3)}{n(n-p-2)},$$

which is zero at  $p = 0$  and strictly positive for  $p \geq 1$  and  $n > p+2$ . With  $p = 2$ , unit residual variances, and balanced arms, the total correction is  $2 \cdot 2 \cdot 5 / \{n(n-4)\} = 20 / \{n(n-4)\}$ .  $\square$

**A.4. Proof of Corollary 1. Corollary 1 (restated; exact cross-population risk).** Under the conditions of Theorem 1 with source rows  $\mathbf{x}_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}_p(\boldsymbol{\mu}_S, \boldsymbol{\Sigma}_S)$ ,  $\boldsymbol{\Sigma}_S \succ \mathbf{0}$ , and any target second-moment matrix  $\mathbf{M}_T$  with  $\mathbf{M}_T[1, 1] = 1$ ,

$$\mathbb{E}[\text{tr}(\mathbf{M}_T \mathbf{S}^{-1})] = a_n + b_n(L_{T|S} - 1), \quad a_n = \frac{1}{n} + \frac{p}{n(n-p-2)}, \quad b_n = \frac{1}{n-p-2},$$

where  $L_{T|S} = \text{tr}(\mathbf{M}_S^{-1} \mathbf{M}_T)$  and  $\mathbf{M}_S = \mathbb{E}_S(\tilde{\mathbf{x}} \tilde{\mathbf{x}}^\top)$ .

*Proof.* Write  $\mathbf{x}_i = \boldsymbol{\mu}_S + \boldsymbol{\Sigma}_S^{1/2} \mathbf{z}_i$  with  $\mathbf{z}_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}_p(\mathbf{0}, \mathbf{I}_p)$ . The augmented rows transform affinely:  $(1, \mathbf{x}_i^\top)^\top = \mathbf{A}(1, \mathbf{z}_i^\top)^\top$  with

$$\mathbf{A} = \begin{pmatrix} 1 & \mathbf{0}^\top \\ \boldsymbol{\mu}_S & \boldsymbol{\Sigma}_S^{1/2} \end{pmatrix}, \quad \mathbf{A}^{-1} = \begin{pmatrix} 1 & \mathbf{0}^\top \\ -\boldsymbol{\Sigma}_S^{-1/2} \boldsymbol{\mu}_S & \boldsymbol{\Sigma}_S^{-1/2} \end{pmatrix},$$

so  $\mathbf{S} = \mathbf{A} \mathbf{S}_z \mathbf{A}^\top$  with  $\mathbf{S}_z$  the standard-normal cross-product, and

$$\text{tr}(\mathbf{M}_T \mathbf{S}^{-1}) = \text{tr}(\mathbf{A}^{-1} \mathbf{M}_T \mathbf{A}^{-\top} \mathbf{S}_z^{-1}) = \text{tr}(\widetilde{\mathbf{M}}_T \mathbf{S}_z^{-1}), \quad \widetilde{\mathbf{M}}_T = \mathbf{A}^{-1} \mathbf{M}_T \mathbf{A}^{-\top}.$$

Two facts about  $\widetilde{\mathbf{M}}_T$ : its  $(1, 1)$  entry equals  $\mathbf{M}_T[1, 1] = 1$ , because the first row of  $\mathbf{A}^{-1}$  is  $(1, \mathbf{0}^\top)$ ; and its trace equals  $L_{T|S}$ , because  $\mathbf{M}_S = \mathbb{E}_S\{\mathbf{A}(1, \mathbf{z}^\top)^\top(1, \mathbf{z}^\top) \mathbf{A}^\top\} = \mathbf{A} \mathbf{A}^\top$ , so  $\text{tr}(\widetilde{\mathbf{M}}_T) = \text{tr}\{\mathbf{M}_T(\mathbf{A} \mathbf{A}^\top)^{-1}\} = \text{tr}(\mathbf{M}_S^{-1} \mathbf{M}_T) = L_{T|S}$ .

It remains to compute  $\mathbb{E}(\mathbf{S}_z^{-1})$ . From the block form (A.5),  $\mathbb{E}\{(\mathbf{S}_z^{-1})_{11}\} = 1/n + p/\{n(n-p-2)\} = a_n$  and  $\mathbb{E}\{(\mathbf{S}_z^{-1})_{22}\} = \mathbb{E}(\mathbf{G}^{-1}) = b_n \mathbf{I}_p$  as in the proof of Theorem 1. The off-diagonal block has expectation  $-\mathbb{E}(\mathbf{G}^{-1} \bar{\mathbf{z}})$ , which vanishes because  $\mathbf{G}$  and  $\bar{\mathbf{z}}$  are independent and  $\mathbb{E}(\bar{\mathbf{z}}) = \mathbf{0}$ . Hence  $\mathbb{E}(\mathbf{S}_z^{-1}) = \text{blockdiag}(a_n, b_n \mathbf{I}_p)$  and

$$\mathbb{E} \text{tr}(\widetilde{\mathbf{M}}_T \mathbf{S}_z^{-1}) = a_n \widetilde{\mathbf{M}}_T[1, 1] + b_n \text{tr}(\widetilde{\mathbf{M}}_T[2:q, 2:q]) = a_n + b_n(L_{T|S} - 1),$$

because the trace of the slope block of  $\widetilde{\mathbf{M}}_T$  is  $\text{tr}(\widetilde{\mathbf{M}}_T) - \widetilde{\mathbf{M}}_T[1, 1] = L_{T|S} - 1$ . The off-diagonal blocks of  $\widetilde{\mathbf{M}}_T$  meet the zero off-diagonal blocks of  $\mathbb{E}(\mathbf{S}_z^{-1})$  and contribute nothing.  $\square$

**Remark A.2 (Consistency checks).** At  $\mathbf{M}_T = \mathbf{M}_S$  we have  $L = q$  and  $a_n + b_n(q-1) = a_n + p b_n = \ell_p(n)$ , recovering Theorem 1. To first order,  $a_n = 1/n + O(n^{-2})$  and  $b_n = 1/n + O(n^{-2})$ , so  $a_n + b_n(L-1) = L/n + O(n^{-2})$ , recovering the first-order statement of Proposition 3. Verification blocks V3e, V3f, and V6c check the block structure and the exact values by independent Monte Carlo over random designs, including a mean-shifted source.

**A.5. Scope.** Two boundaries deserve restating. First, conditional on a realized design, the estimation component  $\sum_a \sigma_{a,S}^2 \text{tr}\{\mathbf{M}_T(\mathbf{S}_a + \mathbf{K}_a)^{-1}\}$  is exact with no distributional assumption on the covariates; Gaussianity enters only when averaging over designs. Second, every statement is on the conditional layer (A6): nothing here integrates over the residual parameters, and no unconditional or full-Bayes version is claimed anywhere in the article.

## Appendix B. Prior Precision: Proof, Misspecification, and the Restricted Illustration

**B.1. Proof of Proposition 1. Proposition 1 (restated).** *Under Assumption 1, suppose the diffuse comparator is proper, so  $\mathbf{S}_a \succ \mathbf{0}$  for each realized arm design. For any  $\mathbf{K}_a \succeq \mathbf{0}$  and any  $\mathbf{M}_T \succeq \mathbf{0}$ , the estimation component of the conditional EPPV is less than or equal to its diffuse-prior value, strictly when  $\mathbf{K}_a \succ \mathbf{0}$  and  $\mathbf{M}_T \succ \mathbf{0}$ . An informative prior may yield a proper posterior when  $\mathbf{S}_a$  is singular, but no diffuse comparison is then defined.*

*Proof.* Fix an arm and a realized design with  $\mathbf{S} \succ \mathbf{0}$  (almost sure under A8 with a flat-prior baseline). From  $\mathbf{K} \succeq \mathbf{0}$  we get  $\mathbf{S} + \mathbf{K} \succeq \mathbf{S}$ , and matrix inversion reverses the Loewner order on the positive definite cone (Horn & Johnson, 2013, Corollary 7.7.4), so

$$(\mathbf{S} + \mathbf{K})^{-1} \preceq \mathbf{S}^{-1}. \quad (\text{B.1})$$

For  $\mathbf{M}_T \succeq \mathbf{0}$ ,

$$\text{tr}\{\mathbf{M}_T \mathbf{S}^{-1}\} - \text{tr}\{\mathbf{M}_T (\mathbf{S} + \mathbf{K})^{-1}\} = \text{tr}\{\mathbf{M}_T^{1/2} (\mathbf{S}^{-1} - (\mathbf{S} + \mathbf{K})^{-1}) \mathbf{M}_T^{1/2}\} \geq 0,$$

because the trace of a positive semidefinite matrix is nonnegative. Averaging over designs preserves the inequality, and summing the two arms gives the claim. Strictness: when  $\mathbf{K} \succ \mathbf{0}$  the difference in (B.1) is positive definite, and when additionally  $\mathbf{M}_T \succ \mathbf{0}$  the trace above is strictly positive.  $\square$

**B.2. Misspecification analysis and the crossing.** Fix one arm and a realized design, write  $\mathbf{V} = (\mathbf{S} + \mathbf{K})^{-1}$ , let  $\mathbf{b} = \boldsymbol{\beta} - \mathbf{m}$  denote the prior-mean offset, and take the residual scale known. Substituting  $\mathbf{y} = \tilde{\mathbf{X}}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  into the posterior mean (A.4) gives

$$\hat{\boldsymbol{\beta}}^{\text{post}} - \boldsymbol{\beta} = -\mathbf{V}\mathbf{K}\mathbf{b} + \mathbf{V}\tilde{\mathbf{X}}^\top \boldsymbol{\varepsilon}, \quad (\text{B.2})$$

so the bias is  $-\mathbf{V}\mathbf{K}\mathbf{b}$  and the sampling covariance is  $\sigma^2\mathbf{V}\mathbf{S}\mathbf{V}$ . The fixed-truth coefficient MSE weighted by  $\mathbf{M}_T$  is

$$\text{MSE}(\mathbf{K}) = \text{tr}[\mathbf{M}_T \{\sigma^2\mathbf{V}\mathbf{S}\mathbf{V} + \mathbf{V}\mathbf{K}\mathbf{b}\mathbf{b}^\top\mathbf{K}\mathbf{V}\}]. \quad (\text{B.3})$$

Specialize to the information-scaled family  $\mathbf{K} = n_0\mathbf{I}_q$  (whitened coordinates) and  $\mathbf{M}_T = \mathbf{I}_q$ , and diagonalize  $\mathbf{S} = \mathbf{U} \text{diag}(s_j)\mathbf{U}^\top$  with  $\tilde{b}_j = (\mathbf{U}^\top\mathbf{b})_j$ . Because  $n_0\mathbf{I}$  commutes with  $\mathbf{S}$ ,

$$\text{MSE}(n_0) = \sum_{j=1}^q \frac{\sigma^2 s_j}{(s_j + n_0)^2} + \sum_{j=1}^q \frac{n_0^2 \tilde{b}_j^2}{(s_j + n_0)^2}. \quad (\text{B.4})$$

The variance sum is strictly decreasing in  $n_0$  and tends to zero; each bias term  $n_0^2 \tilde{b}_j^2 / (s_j + n_0)^2$  is nondecreasing in  $n_0$  and tends to  $\tilde{b}_j^2$ , so  $\text{MSE}(n_0) \rightarrow \|\mathbf{b}\|^2$  as  $n_0 \rightarrow \infty$ .

**Claim (crossing existence).** *If  $\|\mathbf{b}\|^2 > \text{MSE}(0) = \sigma^2 \text{tr}(\mathbf{S}^{-1})$ , there exists  $n_0^\dagger < \infty$  such that  $\text{MSE}(n_0) > \text{MSE}(0)$  for all  $n_0 > n_0^\dagger$ .*

*Proof.*  $\text{MSE}(n_0)$  is continuous on  $[0, \infty)$  by (B.4) and converges to  $\|\mathbf{b}\|^2$  as  $n_0 \rightarrow \infty$ . Since the limit strictly exceeds  $\text{MSE}(0)$ , there is a threshold beyond which the function stays above  $\text{MSE}(0)$ ; continuity on the compact initial segment supplies the finite crossing point by the intermediate value theorem applied to  $\text{MSE}(n_0) - \text{MSE}(0)$ , which is negative for small  $n_0 > 0$  whenever  $\mathbf{b} \neq \mathbf{0}$  is small relative to the variance reduction, and positive in the tail.  $\square$

The two-arm CATE version applies (B.3) per arm and adds the traces, because the CATE-component error is the difference of two independent arm-wise errors. The numbers in Figure B.1(b) come from the verification suite's Monte Carlo over 300 random designs at  $n = 50$ ,  $p = 2$ , with the offset direction  $(0, 1, -1)/\sqrt{2}$  scaled to  $\{0, 0.25, 0.5\}$  residual standard deviations: the half- $\sigma$  path falls from 0.064 at  $n_0 = 0$  to 0.051 near  $n_0 = 10$ , crosses its diffuse value between  $n_0 = 20$  and 35, and reaches 0.186 at  $n_0 = 300$ ; the closed form (B.3) matches an independent noise-level Monte Carlo within 0.4 percent (block V4d).

**B.3. The restricted sample-size illustration.** Under the design-level unit-information prior  $\mathbf{K}_a = n_0\mathbf{M}_S$  (Zellner, 1986), the first-order approximation  $\mathbf{S}_a \approx n_a\mathbf{M}_S$  gives

$$\text{tr}\{\mathbf{M}_S(\mathbf{S}_a + \mathbf{K}_a)^{-1}\} \approx \frac{q}{n_a + n_0}, \quad (\text{B.5})$$

so under balanced arms, homoscedastic residuals, and a common  $n_0$ , the first-order estimation component is  $2\sigma^2 q / (n + n_0)$  and a total-sample-size target shifts down by

$2n_0$ . The display (B.5) replaces a random matrix by its expectation and is a heuristic, not a theorem; outside the three restrictions the displacement is not additive. The main text flags both restrictions.

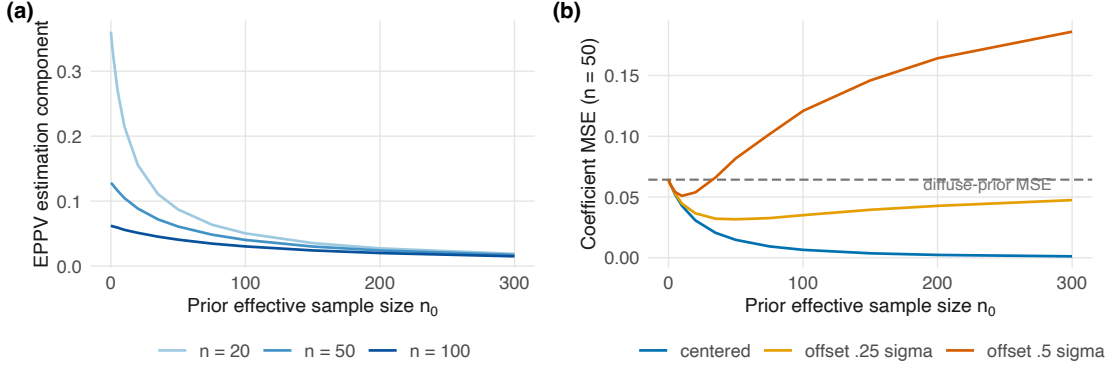


FIGURE B.1. Prior precision as a design lever ( $p = 2$ , information-scaled priors). (a) Conditional-EPPV estimation component against the prior effective sample size  $n_0$  for three arm sizes. (b) Fixed-truth coefficient MSE at  $n = 50$  per arm under correct centering and quarter- and half-residual-standard-deviation offsets; the dashed line marks the diffuse-prior MSE.

## Appendix C. The Shrinkage Family and the Relative Leverage Derivation

**C.1. The family and its endpoints.** Stack the pooled interaction regression as  $\mathbf{y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\varepsilon}$  with  $\mathbf{Z} = [\tilde{\mathbf{X}}, \mathbf{W} \circ \tilde{\mathbf{X}}]$ ,  $\boldsymbol{\theta} = (\boldsymbol{\beta}_0^\top, \boldsymbol{\gamma}^\top)^\top$ ,  $N = n_0 + n_1$ , and common residual scale  $\sigma$ . The path estimator is

$$\hat{\boldsymbol{\theta}}(\lambda) = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \left\{ \|\mathbf{y} - \mathbf{Z}\boldsymbol{\theta}\|^2 + \lambda \|\boldsymbol{\gamma}_{-1}\|^2 \right\} = (\mathbf{Z}^\top \mathbf{Z} + \boldsymbol{\Lambda})^{-1} \mathbf{Z}^\top \mathbf{y}, \quad (\text{C.1})$$

with  $\boldsymbol{\Lambda} = \text{blockdiag}\{\mathbf{0}_{q \times q}, \text{diag}(0, \lambda \mathbf{I}_p)\}$ . Completing the square shows  $\hat{\boldsymbol{\theta}}(\lambda)$  is the posterior mean under a flat prior on  $(\boldsymbol{\beta}_0, \boldsymbol{\gamma}_1)$  and the Gaussian slab  $\boldsymbol{\gamma}_{-1} \sim \mathbf{N}(\mathbf{0}, (\sigma^2/\lambda) \mathbf{I}_p)$ , which is the ridge-prior correspondence used throughout.

The endpoints are derived rather than asserted. At  $\lambda = 0$ , (C.1) is the unrestricted OLS fit of the interaction regression; reparameterizing by arm ( $\boldsymbol{\beta}_1 = \boldsymbol{\beta}_0 + \boldsymbol{\gamma}$ ) shows the fit separates into the two arm-wise regressions, so the  $\boldsymbol{\gamma}$  block equals the difference of the arm-wise OLS coefficients, the unregularized CATE endpoint. As  $\lambda \rightarrow \infty$ , the penalized slopes satisfy  $\hat{\boldsymbol{\gamma}}_{-1}(\lambda) = O(\lambda^{-1})$ , because the normal equations give  $\hat{\boldsymbol{\gamma}}_{-1}(\lambda) = (\mathbf{Z}_2^\top \mathbf{R}_\lambda \mathbf{Z}_2 + \lambda \mathbf{I}_p)^{-1} \mathbf{Z}_2^\top \mathbf{R}_\lambda \mathbf{y}$  for the slope columns  $\mathbf{Z}_2$  and a bounded projection-type matrix  $\mathbf{R}_\lambda$ , and the remaining coordinates converge to the OLS fit of the model

TABLE C.1. Boundary behavior along the shrinkage path (fixed-truth layer;  $p = 2$ ; eight interior penalties per cell; 300 design draws per cell; floor  $\tau^2(0) = 2$  included). The grid oracle is the best of the ten evaluated penalty values, not a continuous optimum. In 36 of the 72 interior cells the interior penalty is worse than the better endpoint.

| Signal    | $n$ | Grid-oracle $\lambda$ | Grid-oracle risk | Best endpoint | Interior worse | Max excess |
|-----------|-----|-----------------------|------------------|---------------|----------------|------------|
| No signal | 20  | $\infty$ (ATE)        | 2.106            | 2.106         | 8 of 8         | 0.221      |
| No signal | 50  | $\infty$ (ATE)        | 2.041            | 2.041         | 8 of 8         | 0.084      |
| No signal | 200 | $\infty$ (ATE)        | 2.010            | 2.010         | 8 of 8         | 0.020      |
| Weak      | 20  | 16                    | 2.190            | 2.235         | 3 of 8         | 0.098      |
| Weak      | 50  | 16                    | 2.092            | 2.129         | 0 of 8         | –          |
| Weak      | 200 | 16                    | 2.028            | 2.030         | 1 of 8         | 0.007      |
| Strong    | 20  | 4                     | 2.275            | 2.367         | 2 of 8         | 0.142      |
| Strong    | 50  | 4                     | 2.115            | 2.128         | 3 of 8         | 0.189      |
| Strong    | 200 | 4                     | 2.030            | 2.030         | 3 of 8         | 0.065      |

without slope interactions, which is the covariate-adjusted ATE endpoint. The path is continuous in  $\lambda$  on  $[0, \infty]$  by continuity of the matrix inverse; the numerical suite represents  $\infty$  by  $\lambda = 10^8$ .

**C.2. Fixed-truth risk and Proposition 2.** Let  $\mathbf{A}_\gamma(\lambda)$  collect the  $\gamma$ -block rows of  $(\mathbf{Z}^\top \mathbf{Z} + \mathbf{\Lambda})^{-1} \mathbf{Z}^\top$ , so  $\hat{\gamma}(\lambda) = \mathbf{A}_\gamma(\lambda) \mathbf{y}$ , and substitute  $\mathbf{y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\varepsilon}$ :

$$R_\gamma(\lambda; d) = \text{tr}[\mathbf{M}_T \{(\mathbb{E}\hat{\gamma} - \gamma)(\mathbb{E}\hat{\gamma} - \gamma)^\top + \sigma^2 \mathbf{A}_\gamma \mathbf{A}_\gamma^\top\}] + \tau_{T|x}^2(\rho_T), \quad \mathbb{E}\hat{\gamma} = \mathbf{A}_\gamma \mathbf{Z}\boldsymbol{\theta}. \quad (\text{C.2})$$

Proposition 2 follows in one line: the infimum over  $\lambda \in [0, \infty]$  is over a set that contains both endpoints, so it is weakly below their minimum. The verified boundary behavior is reported in Table C.1.

**C.3. The EPPV monotonicity obstruction.** Under the flat-plus-slab prior, the posterior covariance of  $\boldsymbol{\theta}$  is  $\sigma^2(\mathbf{Z}^\top \mathbf{Z} + \mathbf{\Lambda})^{-1}$ , so the CATE-component EPPV term is

$$E(\lambda) = \text{tr}[\mathbf{M}_T \sigma^2 \{(\mathbf{Z}^\top \mathbf{Z} + \mathbf{\Lambda})^{-1}\}_{\gamma\gamma}]. \quad (\text{C.3})$$

For  $\lambda' > \lambda$ ,  $\mathbf{\Lambda}' \succeq \mathbf{\Lambda}$ , so by the inversion monotonicity of Appendix B the full inverse decreases in the Loewner order; the  $\gamma\gamma$  principal submatrix inherits the ordering, and the trace against  $\mathbf{M}_T \succeq \mathbf{0}$  makes  $E(\lambda)$  nonincreasing for every realized design and hence in the design average. A criterion that trusts the path’s own prior therefore always weakly prefers more shrinkage: it cannot locate the interior of the fixed-truth frontier, which is the obstruction the tuning half of the article resolves through risk estimation (Proposition 5).

**C.4. Relative leverage: exact and first-order. Proposition 3 (restated).** *Under Assumption 1 with diffuse priors: (a) for Gaussian source rows the estimation component of the target risk is exactly  $\sum_a \sigma_{a,S}^2 \{a_{n_a} + b_{n_a}(L_{T|S} - 1)\}$ , by Corollary 1; (b) to first order it is  $\{\sigma_{0,S}^2/n_0 + \sigma_{1,S}^2/n_1\}L_{T|S}$  with  $L_{T|S} = \text{tr}(\mathbf{M}_S^{-1}\mathbf{M}_T)$ ; and (c) statement (b) extends to non-Gaussian designs under conditions (C1) through (C4) below.*

Part (a) is Corollary 1. For (b), expand  $a_n = 1/n + O(n^{-2})$  and  $b_n = 1/n + O(n^{-2})$ , so  $a_n + b_n(L - 1) = L/n + O(n^{-2})$ . For (c), the conditions are:

- (C1) design rows are i.i.d. from a law with second-moment matrix  $\mathbf{M}_S \succ \mathbf{0}$ ;
- (C2)  $\mathbf{S}_a$  is almost surely nonsingular at every  $n_a \geq q$  in use;
- (C3) the strong law applies coordinatewise,  $\mathbf{S}_a/n_a \rightarrow \mathbf{M}_S$  almost surely;
- (C4) the family  $\{n_a \text{tr}(\mathbf{M}_T \mathbf{S}_a^{-1})\}$  is uniformly integrable.

Under (C1) through (C3), continuity of  $\mathbf{A} \mapsto \text{tr}(\mathbf{M}_T \mathbf{A}^{-1})$  at  $\mathbf{M}_S \succ \mathbf{0}$  gives  $n_a \text{tr}(\mathbf{M}_T \mathbf{S}_a^{-1}) \rightarrow L_{T|S}$  almost surely, and (C4) converts almost-sure convergence into convergence of expectations, which is statement (b). Condition (C4) is not implied by a law of large numbers, because the lower tail of the smallest eigenvalue of  $\mathbf{S}_a/n_a$  controls the inverse trace; the Gaussian frame satisfies (C4) by the exact computation in part (a), and other design laws must be checked. We state (C1) through (C4) explicitly because Assumption 1 alone does not deliver them, even though the limit itself is unsurprising.

Under  $S$ -standardization ( $\mathbf{M}_S = \mathbf{I}_q$ ), the leverage decomposes as

$$L_{T|S} = \text{tr}(\mathbf{M}_T) = 1 + \|\boldsymbol{\mu}_T\|^2 + \text{tr}(\boldsymbol{\Sigma}_T) = 1 + M_{T|S} + D_{T|S},$$

matching the mean-shift and dispersion terms of the anchor. Two boundary facts close the appendix. No bound  $L_{T|S} \geq q$  holds: the overdispersed scenario with source standard deviation 2 in each of  $p = 2$  directions gives  $L = 1 + 2/4 = 1.5 < 3$ . And leverage multiplies the source residual variances, so a low-leverage noisy source can lose on total risk: with  $\sigma_{a,S}^2 = 4$  and leverage 1.5 against a representative design with  $\sigma_{a,S}^2 = 1$  and leverage 3, the first-order totals at  $n = 20$  are 2.60 against 2.30, and the exact totals from Corollary 1 are 2.70 against 2.36 (blocks V6b, V6c).

## Appendix D. The Unidentified Correlation: Proofs and the Coverage Study

**D.1. Proof of Remark 1.** Each unit reveals  $y_i = Y_i(W_i)$  and conceals  $Y_i(1 - W_i)$ . Randomization makes  $W_i$  independent of the potential outcomes given  $\mathbf{x}_i$ , so the

density of the observed data integrates the joint bivariate Gaussian residual law over the concealed coordinate. The marginal of one coordinate of a bivariate Gaussian depends only on that coordinate's mean and variance, so

$$p(\mathbf{y}^{\text{obs}} \mid \boldsymbol{\beta}_0, \boldsymbol{\beta}_1, \Sigma_\varepsilon) = \prod_{i: W_i=0} \mathcal{N}(y_i; \tilde{\mathbf{x}}_i^\top \boldsymbol{\beta}_0, \sigma_{0,S}^2) \prod_{i: W_i=1} \mathcal{N}(y_i; \tilde{\mathbf{x}}_i^\top \boldsymbol{\beta}_1, \sigma_{1,S}^2), \quad (\text{D.1})$$

which contains the two variances but not  $\rho$ . The likelihood is therefore constant in  $\rho$ , and under a prior in which  $\rho$  is a priori independent of the remaining parameters, the marginal posterior of  $\rho$  equals its prior. Treating  $\rho$  as learned from a randomized experiment is circular; the classical references are cited in the main text.

**D.2. Proofs of Proposition 4 and Corollary 2.** Proposition 4: if candidates share the target, the model, and the marginal residual scales, then  $R_c(d; \rho) - R_c(d'; \rho) = E(d) - E(d')$  is free of  $\rho$ , which is the claim. The restriction does the work: a candidate set that changes the covariate list, the working model, or the target changes the floor, and rankings can then rotate with  $\rho$ .

Corollary 2: on the first-order layer,  $R_c(n; \rho) = 2\sigma^2 q/n + \tau_{T|x}^2(\rho)$  is continuous and strictly decreasing in  $n$  with limit  $\tau_{T|x}^2(\rho)$ , so  $R_c(n; \rho) \leq R^*$  has a finite solution if and only if the margin  $R^* - \tau_{T|x}^2(\rho)$  is positive, and solving gives  $n^*(\rho) = 2\sigma^2 q / \{R^* - \tau_{T|x}^2(\rho)\}$ . The threshold is real-valued; an implementable size is the smallest integer above it, subject to the domain requirement  $n > p + 2$  of the exact layer.

**Claim (the exact requirement exceeds the first-order requirement).** *Let  $n_{\text{ex}}^*$  solve  $2\sigma^2 \ell_p(n) = R^* - \tau^2$  and  $n_{\text{fo}}^*$  solve  $2\sigma^2 q/n = R^* - \tau^2$ , both on  $n > p + 2$ . Then  $n_{\text{ex}}^* \geq n_{\text{fo}}^*$ , strictly for  $p \geq 1$ .*

*Proof.* By Theorem 1,  $\ell_p(n) - q/n = p(p+3)/\{n(n-p-2)\} \geq 0$  with strict inequality for  $p \geq 1$ , so  $2\sigma^2 \ell_p(n) \geq 2\sigma^2 q/n$  at every  $n$  in the domain. Both maps are continuous and strictly decreasing to zero on  $(p+2, \infty)$  (for  $\ell_p$ , both summands decrease). At  $n = n_{\text{fo}}^*$  the exact risk therefore still exceeds the margin, so the exact crossing happens at a larger  $n$ .  $\square$

The margin-to-size table in the main text (Table 2) evaluates both requirements at  $R^* = 2.25$ : first-order  $\{\infty, 24.0, 7.06, 4.14, 2.93\}$  and exact  $\{\infty, 27.4, 10.6, 7.8, 6.6\}$  across  $\rho \in \{-0.5, 0, 0.3, 0.6, 0.9\}$  (block V7).

**D.3. Coverage-study design.** Each replication draws fresh designs of  $n$  units per arm with standardized Gaussian moderators ( $p = 2$ ); truths  $\boldsymbol{\beta}_0 = (0, 0.4, 0.2)^\top$  and  $\boldsymbol{\gamma} = (0.3, 0.25, -0.25)^\top$  with unit residual scales; computes the arm-wise conjugate



posteriors at prior effective sample size  $n_0$  (precision  $n_0 \mathbf{I}_q$ ), with the treated-arm prior mean optionally offset by half a residual standard deviation in the slope directions  $(0, 1, -1)/\sqrt{2}$ ; draws a new target unit  $\tilde{\mathbf{x}}$ ; and checks whether the realized USTE  $\delta = \tilde{\mathbf{x}}^\top \boldsymbol{\gamma} + \nu$ ,  $\nu \sim \mathbf{N}\{0, \tau^2(\rho_{\text{true}})\}$ , falls in

$$\tilde{\mathbf{x}}^\top (\hat{\boldsymbol{\beta}}_1^{\text{post}} - \hat{\boldsymbol{\beta}}_0^{\text{post}}) \pm 1.96 \sqrt{\tilde{\mathbf{x}}^\top (\mathbf{V}_1 + \mathbf{V}_0) \tilde{\mathbf{x}} + \tau^2(\rho_{\text{assumed}})}, \quad \mathbf{V}_a = (\mathbf{S}_a + \mathbf{K}_a)^{-1}. \quad (\text{D.2})$$

The study uses 4,000 replications per cell with  $\rho_{\text{true}} = 0.3$  throughout, the prior grid  $n \in \{20, 100\} \times n_0 \in \{0, 5, 20, 50\} \times \text{offset} \in \{0, 0.5\}\sigma$ , and the correlation grid  $\rho_{\text{assumed}} \in \{-0.5, 0, 0.3, 0.6, 0.9\}$  under diffuse priors. Results (blocks V16a through V16c): nominal-case coverage .950 to .960; assumed 0.6 covers at .892 and .865; assumed 0.9 covers at .713 and .592; assumed values below the truth cover at .976 to .996. The asymmetry follows from the floor: an assumed correlation above the truth shrinks the interval's  $\tau$  term below the realized heterogeneity.

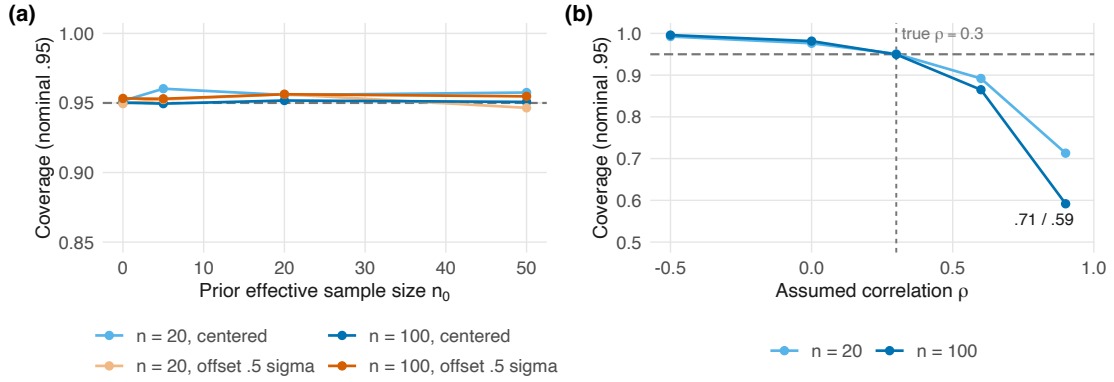


FIGURE D.1. Empirical coverage of nominal 95% posterior-predictive USTE intervals (4,000 replications per cell). (a) Coverage against prior effective sample size, with and without a treated-arm prior-mean offset, with  $\rho = 0.3$  correctly specified. (b) Coverage against the assumed correlation under diffuse priors; the dotted line marks the true value 0.3.

## Appendix E. The Canonical Regime and the Tuning Proofs

**E.1. Proof of Lemma 2. Lemma 2 (restated).** *Under matched target moments, a balanced two-arm design of  $n$  per arm, source-whitened moderators, and common known scale  $\sigma$ : conditional on the realized design the OLS slope contrast is exactly Gaussian with covariance  $\sigma^2$  times the slope block of  $(\mathbf{Z}^\top \mathbf{Z})^{-1}$ ; unconditionally*

$\sqrt{n}(\hat{\gamma}_{-1} - \gamma_{-1}) \Rightarrow \mathbf{N}_p(\mathbf{0}, 2\sigma^2 \mathbf{I}_p)$ ; and the slope-block fixed-truth USTE risk of any rule is  $\mathbb{E}\|\tilde{\gamma}_{-1} - \gamma_{-1}\|^2$  up to terms common to all rules on the path.

*Proof.* Conditional exactness is the standard Gaussian linear-model fact. For the limit, write  $\mathbf{Z}^\top \mathbf{Z}$  in the arm blocks

$$\mathbf{Z}^\top \mathbf{Z} = \begin{pmatrix} \mathbf{S} & \mathbf{S}_1 \\ \mathbf{S}_1 & \mathbf{S}_1 \end{pmatrix}, \quad \mathbf{S} = \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}, \quad \mathbf{S}_1 = \sum_{i:W_i=1} \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i^\top,$$

so the  $\gamma$  block of the inverse is  $(\mathbf{S}_1 - \mathbf{S}_1 \mathbf{S}^{-1} \mathbf{S}_1)^{-1}$  by block inversion. With i.i.d. whitened rows of finite fourth moment and balanced arms, the law of large numbers and the central limit theorem give  $\mathbf{S} = N \mathbf{I}_q + O_p(\sqrt{N})$  and  $\mathbf{S}_1 = (N/2) \mathbf{I}_q + O_p(\sqrt{N})$  with  $N = 2n$ , hence

$$\mathbf{S}_1 - \mathbf{S}_1 \mathbf{S}^{-1} \mathbf{S}_1 = \frac{N}{2} \mathbf{I}_q - \frac{N}{2} \mathbf{I}_q (N \mathbf{I}_q)^{-1} \frac{N}{2} \mathbf{I}_q + O_p(\sqrt{N}) = \frac{N}{4} \mathbf{I}_q + O_p(\sqrt{N}),$$

so the  $\gamma$ -block covariance is  $\sigma^2(4/N) \mathbf{I}_q + o_p(N^{-1})$ , whose slope subblock gives  $v = 2\sigma^2/n$ ; Slutsky's theorem converts the design-conditional exact normal law into the stated limit. Averaging the exact conditional covariance over random designs yields a covariance mixture, not an exact spherical normal, which is why the unconditional spherical display is asymptotic; the finite-design track prices the finite- $n$  difference. For the loss reduction: under matched target moments ( $\mathbb{E}_T \mathbf{x} = \mathbf{0}$ ,  $\mathbb{E}_T \mathbf{x} \mathbf{x}^\top = \mathbf{I}_p$ ), the target-integrated squared error of the CATE component separates into the average-effect coordinate, the slope block in Euclidean norm, and the floor; every rule on the path leaves the average-effect coordinate at its OLS value and no rule touches the floor, so both are common additive terms and the slope block carries every comparison. A ridge penalty  $\lambda$  on the slopes multiplies the canonical slope estimate by  $c = (N/4)/\{(N/4) + \lambda\}$ , sweeping  $[0, 1]$ ; this is the penalty-to-factor map used in the main text.  $\square$

Rotation invariance is exact, not approximate: every rule in Table 3 of the main text except cross-validation multiplies  $\hat{\gamma}_{-1}$  by a scalar function of  $\|\hat{\gamma}_{-1}\|^2$ , so canonical risks depend on  $\gamma_{-1}$  only through  $G = \|\gamma_{-1}\|^2$  and the  $\eta$  grid is exhaustive.

**E.2. Proof of Theorem 2.** With  $\mathbf{X} = \hat{\gamma}_{-1} \sim \mathbf{N}_p(\gamma, v \mathbf{I}_p)$  and  $\delta_c = c \mathbf{X}$ ,

$$R(c) = \mathbb{E}\|c \mathbf{X} - \gamma\|^2 = c^2 \mathbb{E}\|\mathbf{X} - \gamma\|^2 + (1 - c)^2 \|\gamma\|^2 = c^2 p v + (1 - c)^2 G,$$

because the cross term has mean zero. Minimizing the quadratic gives  $c^{\text{lin}} = G/(G + pv)$  and  $R(c^{\text{lin}}) = pvG/(G + pv) = pv\eta/(1 + \eta)$ . The gain over the better endpoint is  $h(\eta) = pv\{\min(1, \eta) - \eta/(1 + \eta)\}$ ; on  $\eta \leq 1$ ,  $h/pv = \eta^2/(1 + \eta)$  is increasing, and

on  $\eta \geq 1$ ,  $h/pv = 1/(1 + \eta)$  is decreasing, so the maximum sits at the kink  $\eta = 1$  with value  $pv/2$ , and the limits at 0 and  $\infty$  are zero. The factor  $c^{\text{lin}}$  is the best *fixed linear* factor only; no claim is made against nonlinear or data-dependent rules, and the simulation reports excesses relative to this comparator without treating it as a bound.

**E.3. Proof of Theorem 3.** *Part (i).* For  $\delta_b(\mathbf{X}) = \mathbf{X} - bv\mathbf{X}/\|\mathbf{X}\|^2$ ,

$$R(\delta_b; \gamma) = pv - 2bv \mathbb{E} \left[ \frac{(\mathbf{X} - \gamma)^\top \mathbf{X}}{\|\mathbf{X}\|^2} \right] + b^2 v^2 \mathbb{E} \left[ \frac{1}{\|\mathbf{X}\|^2} \right].$$

Stein's identity (Stein, 1981) with  $\mathbf{g}(\mathbf{x}) = \mathbf{x}/\|\mathbf{x}\|^2$ , whose divergence is  $(p-2)/\|\mathbf{x}\|^2$ , is applicable for  $p \geq 3$  because the inverse second moment of a noncentral chi-square with at least three degrees of freedom is finite, and it gives

$$R(\delta_b; \gamma) = pv - v^2 b \{2(p-2) - b\} \mathbb{E} \left[ \frac{1}{\|\mathbf{X}\|^2} \right],$$

strictly below  $pv$  for every  $\gamma$  exactly when  $0 < b < 2(p-2)$ , and equal to  $pv$  at the boundary  $b = 2(p-2)$ .

*Part (ii).* Both rules have the form  $h(T)\mathbf{X}$  with  $T = \|\mathbf{X}\|^2$ ,  $h(t) = 1 - bv/t$ , and  $h_+ = \max(h, 0)$ . Then

$$R(h\mathbf{X}) - R(h_+\mathbf{X}) = \mathbb{E}[\{h^2(T) - h_+^2(T)\}T] - 2\mathbb{E}[\{h(T) - h_+(T)\}\mathbf{X}^\top \gamma].$$

On  $\{h < 0\}$  the first integrand is  $h^2 T > 0$ ; elsewhere both vanish. For the second term,  $h - h_+ \leq 0$  and  $\mathbb{E}(\mathbf{X}^\top \gamma \mid T = t) \geq 0$  for every  $t$ , because on the sphere of radius  $\sqrt{t}$  the conditional density is proportional to  $\exp(\mathbf{x}^\top \gamma/v)$  and weights each point more than its antipode when  $\mathbf{x}^\top \gamma > 0$  (Baranchik, 1970; Lehmann & Casella, 1998). Both terms are therefore nonnegative and the first is strictly positive whenever  $\mathbb{P}\{h(T) < 0\} > 0$ , which holds at every  $\gamma$  because  $T$  has full support on  $(0, \infty)$ . The constants follow:  $b = p - 2$  lies in the window for  $p \geq 3$ ;  $b = p$  lies in the window for  $p \geq 5$ , sits exactly on the boundary at  $p = 4$  (plain risk  $\equiv pv$ , so the strict positive-part improvement alone delivers strict dominance), and lies outside it at  $p = 3$ , where dominance in fact fails: deterministic quadrature exhibits risks  $3.19v$  and  $3.13v$  at  $\eta = 4$  and 8 against the endpoint's  $3v$  (block V10a).

*Part (iii).* At  $\gamma = \mathbf{0}$ ,  $T/v \sim \chi_p^2$  and  $\mathbb{E}[v/T] = 1/(p-2)$ , so with  $b = p-2$  the plain risk is  $pv - (p-2)^2 v/(p-2) = 2v$ , independent of  $p$ ; the positive-part rule is strictly better by part (ii). At  $p = 4$  with  $b = 4$  the positive-part risk at the origin has the

closed form  $v[4 + \mathbb{E}\{(Z - 8)\mathbf{1}(Z \leq 4)\}]$  with  $Z \sim \chi_4^2$ , whose value is  $0.5413v$  (block V10c); the display follows by evaluating  $\mathbb{E}[h_+^2 T]$  on the truncation event.

*Part (iv).* For  $p \leq 2$  the inverse-norm expectation diverges (the chi-square density with at most two degrees of freedom is bounded away from zero near the origin), so every plain rule with  $b \neq 0$  has infinite risk; the deterministic capped-moment computation in block V11 exhibits the divergence. And no rule at all is weakly better than the CATE endpoint at every truth, because  $\mathbf{X}$  is admissible in one and two dimensions under Euclidean loss (Lehmann & Casella, 1998).

**E.4. Proof of Theorem 4 (target-weighted loss).** Throughout,  $p \geq 3$ ,  $\mathbf{W} \succ \mathbf{0}$ ,  $T = \|\mathbf{X}\|^2$ , and the rule is  $\delta_b = h(T)\mathbf{X}$  with  $h(t) = 1 - bv/t$ ; the factor uses the Euclidean norm while the loss is weighted, and this mismatch is what the theorem addresses. By rotation of coordinates we may take  $\mathbf{W} = \text{diag}(w_1, \dots, w_p)$  with  $w_1 = \lambda_{\max}$ .

*Part (i): the identity.* Expanding the weighted loss,

$$R_{\mathbf{W}}(\delta_b; \gamma) = \mathbb{E}[(\mathbf{X} - \gamma)^\top \mathbf{W}(\mathbf{X} - \gamma)] - 2bv \mathbb{E}\left[\frac{(\mathbf{X} - \gamma)^\top \mathbf{W} \mathbf{X}}{T}\right] + b^2 v^2 \mathbb{E}\left[\frac{\mathbf{X}^\top \mathbf{W} \mathbf{X}}{T^2}\right].$$

The first term is  $v \text{tr}(\mathbf{W})$ . For the middle term apply Stein's identity coordinatewise with  $g_j(\mathbf{x}) = (\mathbf{W}\mathbf{x})_j / \|\mathbf{x}\|^2$ :

$$\sum_j \frac{\partial g_j}{\partial x_j} = \frac{\text{tr}(\mathbf{W})}{\|\mathbf{x}\|^2} - \frac{2\mathbf{x}^\top \mathbf{W} \mathbf{x}}{\|\mathbf{x}\|^4},$$

so  $\mathbb{E}[(\mathbf{X} - \gamma)^\top \mathbf{W} \mathbf{X} / T] = v \mathbb{E}[\text{tr}(\mathbf{W}) / T - 2\mathbf{X}^\top \mathbf{W} \mathbf{X} / T^2]$ , and collecting terms gives

$$R_{\mathbf{W}}(\delta_b; \gamma) = v \text{tr}(\mathbf{W}) + v^2 \mathbb{E}\left[-\frac{2b \text{tr}(\mathbf{W})}{T} + \frac{(4b + b^2) \mathbf{X}^\top \mathbf{W} \mathbf{X}}{T^2}\right]. \quad (\text{E.1})$$

Integrability for  $p \geq 3$  follows from  $\mathbf{X}^\top \mathbf{W} \mathbf{X} \leq \lambda_{\max} T$  and  $\mathbb{E}[1/T] < \infty$ .

*Part (ii): the window.* Bounding  $\mathbf{X}^\top \mathbf{W} \mathbf{X} \leq \lambda_{\max} T$  in (E.1),

$$R_{\mathbf{W}}(\delta_b; \gamma) - v \text{tr}(\mathbf{W}) \leq v^2 \lambda_{\max} \mathbb{E}\left[\frac{1}{T}\right] b \{b - 2(p_{\text{eff}} - 2)\}, \quad p_{\text{eff}} = \frac{\text{tr}(\mathbf{W})}{\lambda_{\max}},$$

which is strictly negative for  $0 < b < 2(p_{\text{eff}} - 2)$  and nonpositive at the boundary  $b = 2(p_{\text{eff}} - 2) > 0$ . The boundary statement is sharp: if  $\mathbf{W} = w\mathbf{I}_p$ , then  $p_{\text{eff}} = p$ , the matrix bound is equality, and the plain rule at  $b = 2(p - 2)$  has risk exactly equal to the endpoint at every truth. The condition  $p_{\text{eff}} > 2$  is therefore an

existence condition for some positive  $b$ , not a certificate for a prespecified menu constant; this is the Bock effective-dimension condition (Bock, 1975), with the arbitrary-quadratic-loss antecedent in Berger (1976). For the positive-part improvement under weighted loss, the same two-term comparison as in Theorem 3(ii) applies: the first term  $\mathbb{E}[\{h^2 - h_+^2\} \mathbf{X}^\top \mathbf{W} \mathbf{X}]$  is strictly positive on  $\{h < 0\}$  because  $\mathbf{X}^\top \mathbf{W} \mathbf{X} > 0$  almost surely, and the cross term is controlled by the index-shift identity  $\mathbb{E}[\gamma_j X_j g(T)] = \gamma_j^2 \mathbb{E}[g(T^{(2)})]$  for nonnegative  $g$ , where  $T^{(2)}/v \sim \chi_{p+2}^2(\lambda)$  with  $\lambda = \|\boldsymbol{\gamma}\|^2/v$ ; summing over coordinates gives  $\mathbb{E}[\mathbf{X}^\top \mathbf{W} \boldsymbol{\gamma} g(T)] = (\sum_j w_j \gamma_j^2) \mathbb{E}[g(T^{(2)})] \geq 0$ , so the positive part strictly improves the plain rule at every truth. Substituting  $b = p - 2$  gives the JS+ requirement  $p_{\text{eff}} \geq (p + 2)/2$ ; substituting  $b = p$  gives the EB/SURE+ requirement  $p_{\text{eff}} \geq (p + 4)/2$ . Equality in either threshold uses the weak plain-rule boundary followed by strict positive-part improvement.

*Part (iii): the origin premium.* At  $\boldsymbol{\gamma} = \mathbf{0}$ , the index-shift identities give  $\mathbb{E}[X_j^2 g(T)] = v \mathbb{E}[g(T^{(2)})]$  with  $T^{(2)}/v \sim \chi_{p+2}^2$  central, so

$$R_{\mathbf{W}}(\delta_b; \mathbf{0}) = v \operatorname{tr}(\mathbf{W}) \mathbb{E}[h(T^{(2)})^2] = v \operatorname{tr}(\mathbf{W}) \left\{ 1 - \frac{2b}{p} + \frac{b^2}{p(p-2)} \right\},$$

using  $\mathbb{E}[v/T^{(2)}] = 1/p$  and  $\mathbb{E}[v^2/(T^{(2)})^2] = 1/\{p(p-2)\}$ . At  $b = p - 2$  the brace evaluates to  $1 - 2(p-2)/p + (p-2)/p = 2/p$ , giving the stated premium  $2v \operatorname{tr}(\mathbf{W})/p$  exactly (block V13c confirms to  $10^{-6}$ ).

*Part (iv): the empty window.* Suppose  $p_{\text{eff}} \leq 2$  and fix  $b > 0$ . Take  $\boldsymbol{\gamma} = m\mathbf{u}_1$  along the top eigendirection and let  $\lambda = m^2/v \rightarrow \infty$ . We claim

$$\frac{m^2}{v^2} \{R_{\mathbf{W}}(\delta_b; \boldsymbol{\gamma}) - v \operatorname{tr}(\mathbf{W})\} \longrightarrow \lambda_{\max} b \{b + 4 - 2p_{\text{eff}}\} > 0, \quad (\text{E.2})$$

the right side being positive because  $b > 0$  and  $4 - 2p_{\text{eff}} \geq 0$ . By (E.1) it suffices to show  $m^2 \mathbb{E}[1/T] \rightarrow 1$  and  $m^2 \mathbb{E}[\mathbf{X}^\top \mathbf{W} \mathbf{X}/T^2] \rightarrow \lambda_{\max}$ .

For the first limit,  $T/v \sim \chi_p^2(\lambda)$  has the Poisson-mixture representation  $\chi_{p+2K}^2$  with  $K \sim \text{Poisson}(\lambda/2)$ , and  $\mathbb{E}[1/\chi_{p+2K}^2 \mid K] = 1/(p + 2K - 2)$  for  $p \geq 3$ . Then  $m^2 \mathbb{E}[1/T] = \lambda \mathbb{E}[1/(p + 2K - 2)]$ , and  $\lambda/(p + 2K - 2) \rightarrow 1$  almost surely while, on the event  $\{K \geq \lambda/4\}$ , the ratio is at most  $\lambda/(\lambda/2 + p - 2) \leq 2$ , and the complementary event has probability at most  $e^{-c\lambda}$  on which the ratio is at most  $\lambda/(p - 2)$ ; dominated convergence in this split form gives the limit 1.

For the second limit, write  $\mathbf{X} = (X_1, \mathbf{X}_{-1})$  in the eigenbasis,  $Q = \|\mathbf{X}_{-1}\|^2$ , so  $\mathbf{X}^\top \mathbf{W} \mathbf{X} = \lambda_{\max} X_1^2 + \sum_{j \geq 2} w_j X_j^2$  and  $T = X_1^2 + Q$ . The upper bound  $\mathbf{X}^\top \mathbf{W} \mathbf{X} \leq \lambda_{\max} T$  gives  $\limsup m^2 \mathbb{E}[\cdot] \leq \lambda_{\max}$  by the first limit. For the matching lower bound it suffices that  $m^2 \mathbb{E}[Q/T^2] \rightarrow 0$ , because  $\mathbb{E}[X_1^2/T^2] = \mathbb{E}[1/T] - \mathbb{E}[Q/T^2]$ . Split on  $A =$

$\{|X_1| \geq m/2\}$ . On  $A$ ,  $T^2 \geq (m^2/4)^2$ , so  $\mathbb{E}[Q/T^2 \mathbf{1}_A] \leq 16(p-1)v/m^4$ , whose product with  $m^2$  vanishes. On  $A^c$ , the Gaussian density of  $X_1 \sim \mathcal{N}(m, v)$  on  $\{|x| \leq m/2\}$  is bounded by  $\phi(m/(2\sqrt{v}))/\sqrt{v} \leq Ce^{-m^2/(8v)}$ , so, using independence of  $X_1$  and  $Q$  and  $\int_{|x| \leq m/2} (x^2 + Q)^{-1} dx \leq \pi Q^{-1/2}$ ,

$$\mathbb{E}\left[\frac{Q}{T^2} \mathbf{1}_{A^c}\right] \leq \mathbb{E}\left[\frac{1}{X_1^2 + Q} \mathbf{1}_{A^c}\right] \leq Ce^{-m^2/(8v)} \{\pi \mathbb{E}[Q^{-1/2}] + m\},$$

and  $\mathbb{E}[Q^{-1/2}] < \infty$  for  $p-1 \geq 2$ ; the product with  $m^2$  vanishes. This proves (E.2): for every fixed  $b > 0$  there are truths at which the plain rule's weighted risk strictly exceeds the CATE endpoint's, so no member of the family is weakly better than the endpoint for all truths. Deterministic quadrature exhibits the excess at finite signal (block V13d: excesses 0.10 to 2.09 at  $m = 10$  across  $b \in \{0.5, 1, 2, 4\}$  for the  $p_{\text{eff}} = 1.15$  shape). The statement concerns the Euclidean-factor family; it does not assert that no rule of any form improves on the endpoint under weighted loss, and rules whose factor uses the  $\mathbf{W}$ -geometry are outside this article's menu.

**E.5. Proof of Theorem 5.** If  $R(\tilde{\gamma}; \mathbf{0}) = \mathbb{E}_0 \|\tilde{\gamma}(\mathbf{X})\|^2 = 0$ , then  $\tilde{\gamma}(\mathbf{X}) = \mathbf{0}$  almost surely under  $\mathcal{N}_p(\mathbf{0}, v\mathbf{I})$ . For  $v > 0$  this Gaussian measure and Lebesgue measure are mutually absolutely continuous, so the set  $\{\tilde{\gamma} \neq \mathbf{0}\}$  is Lebesgue-null and hence null under  $\mathcal{N}_p(\gamma, v\mathbf{I})$  for every  $\gamma$ : the rule equals the ATE endpoint up to almost-everywhere equivalence, and its risk function is  $G$ . A rule weakly better than both endpoints at all truths would need risk 0 at  $\gamma = \mathbf{0}$  (matching the ATE endpoint) and hence would be the ATE endpoint, which is strictly worse than the CATE endpoint at any truth with  $G > pv$ ; contradiction. The premium statement is the contrapositive. The proof uses nothing about the path, linearity, or rotation invariance, so the impossibility binds every measurable procedure in the regime.

**E.6. Proof of Proposition 5 and the selection caveat.** For fixed  $c$ ,  $\mathbb{E}\|\mathbf{X}\|^2 = G + pv$  gives

$$\begin{aligned} \mathbb{E}\{\text{SURE}(c)\} &= (1-c)^2(G + pv) + (2c-1)pv \\ &= (1-c)^2G + pv\{(1-c)^2 + 2c-1\} \\ &= (1-c)^2G + c^2pv = R(c). \end{aligned}$$

The minimizer over  $c \in [0, 1]$ : the quadratic's stationary point is  $c^\circ = 1 - pv/\|\mathbf{X}\|^2 < 1$ , so projection truncates only at zero and  $\hat{c} = (1 - pv/\|\mathbf{X}\|^2)_+$ , which is the empirical-Bayes moment factor  $\hat{G}_+ / (\hat{G}_+ + pv)$ ; the identity is noted in the homoscedastic hierarchical setting by Xie et al. (2012).

For the selection statement, put  $T = \|\mathbf{X}\|^2$  and  $a = pv$ . The fixed- $c$  expression evaluated at  $\hat{c}$  is the pointwise minimum. On  $T \leq a$ , both the selected estimator  $\delta(\mathbf{X}) = \mathbf{0}$  and the minimum have SURE  $T - pv$ . On  $T > a$ ,  $\delta(\mathbf{X}) = (1 - a/T)\mathbf{X}$  and

$$\text{div } \delta(\mathbf{X}) = p - (p - 2)a/T.$$

Stein's formula for this data-dependent estimator is  $\|\delta - \mathbf{X}\|^2 + 2v \text{div } \delta - pv$ . Subtracting the pointwise fixed- $c$  minimum gives exactly  $4pv^2/T$  on  $T > a$  and zero otherwise. Taking expectations yields

$$\mathbb{E}\{\min_c \text{SURE}(c)\} = R(\delta) - 4pv^2 \mathbb{E}\{T^{-1} \mathbf{1}(T > pv)\}.$$

The correction is finite and strictly positive for every truth because the noncentral chi-square law has positive mass above  $pv$ . Thus the minimized fixed- $c$  SURE is strictly optimistic for the selected rule; reporting its risk requires cross-fitting, an independent split, or a selection-adjusted estimator.

**E.7. The hierarchical-Bayes specification.** With  $\boldsymbol{\gamma}_{-1} \mid g \sim \mathbf{N}_p(\mathbf{0}, g\mathbf{I}_p)$ , the marginal is  $\mathbf{X} \mid g \sim \mathbf{N}_p(\mathbf{0}, (g + v)\mathbf{I}_p)$  and the posterior-mean factor is

$$\bar{c}(\mathbf{X}) = \mathbb{E}\left\{\frac{g}{g + v} \mid \mathbf{X}\right\} = \frac{\int \frac{g}{g + v} (g + v)^{-p/2} \exp\{-\|\mathbf{X}\|^2/2(g + v)\} \pi(dg)}{\int (g + v)^{-p/2} \exp\{-\|\mathbf{X}\|^2/2(g + v)\} \pi(dg)}, \quad (\text{E.3})$$

a function of  $\|\mathbf{X}\|^2$  alone. The hyperprior adopted here is proper and elicited on an interpretable marginal scale. Write  $s = \sqrt{g} = A|Z|$ , where  $Z \sim \mathbf{N}(0, 1)$ , and let  $Q \sim \chi_p^2$  independently. Since  $\boldsymbol{\gamma}_{-1} \mid g \sim \mathbf{N}_p(\mathbf{0}, g\mathbf{I}_p)$ ,

$$\eta = \frac{\|\boldsymbol{\gamma}_{-1}\|^2}{pv} = \frac{A^2 Z^2 Q}{pv}.$$

We solve  $\mathbb{P}(A^2 Z^2 Q/(pv) > 1) = 1/2$  by adaptive integration and root finding. Thus  $A = a_p \sqrt{v}$ , with  $a_p = 2.040279, 1.689291$ , and  $1.582074$  for  $p = 2, 5$ , and  $10$ , respectively; substituting each root returns probability 0.5 to numerical tolerance. This calibrates the marginal signal distribution, not merely the conditional variance  $g$ . The posterior factor in (E.3) is evaluated by adaptive quadrature on  $s$  with convergence diagnostics and a monotone spline interpolant in  $\log \|\mathbf{X}\|^2$ ; block V14 checks calibration, direct-versus-interpolated factors, and integration error separately. No dominance property is claimed for the rule (Gelman, 2006).

A log-uniform hyperprior on  $g/v$  truncated to  $[10^{-4}, 10^4]$  is a natural alternative specification. Its zero-signal risk is determined by the lower truncation point rather than by the data: recomputing at lower cutoffs  $10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}$  gives zero-signal

risks 0.401, 0.203, 0.137, 0.104 in units of  $v$  (documented in block V14). We therefore do not use it. Under the marginally calibrated half-normal prior at  $p = 5$ , multiplying  $A$  by 0.5, 1, and 2 gives zero-signal premiums  $0.386v$ ,  $0.712v$ , and  $0.897v$  (block V14). The ratio 2.32 is not negligible, so Figure F.2 reports the full sensitivity rather than treating the hyperprior as stable.

## Appendix F. Computational Verification and Simulation Settings

**F.1. Evidence taxonomy.** Every verification block carries one of five evidence labels, and the map below states the label next to each block, so that the strength of evidence behind each block is explicit:

- **proof:** the claim is proved in Appendices A through E; the block is a consistency companion;
- **quadrature:** deterministic noncentral chi-square or checked one-dimensional quadrature assertion, validated to the reported numerical tolerance and carrying no Monte Carlo error;
- **independent:** an independent numerical derivation (Monte Carlo or optimization against a separately derived closed form);
- **regression:** a same-implementation regression or transcription check, useful against coding drift but not independent validation;
- **descriptive:** a reported number with no assertion attached.

The canonical risks of every spherical-factor rule, including the hierarchical rule, are computed by the index-shift quadrature described in Appendix E, which replaces Monte Carlo as the primary canonical evidence; a 200,000-replication Monte Carlo agreement track runs alongside (worst relative deviation 0.013; block V15).

### F.2. Verification map.

TABLE F.1. Verification blocks (suite `01_verification.R`, seed 20260801; zero failures on the shipped run).

| Block | Evidence    | Statement checked                                                                                           |
|-------|-------------|-------------------------------------------------------------------------------------------------------------|
| V1    | independent | Lemma 1 closed form against a brute-force joint-posterior build (diff 0)                                    |
| V1b   | regression  | Conjugate posterior mean against a numerical posterior mode; predictive variance against posterior sampling |
| V2    | regression  | First-order diffuse EPPV equals the benchmark on a 25-cell grid (transcription check)                       |



| Block       | Evidence               | Statement checked                                                                                                                                                                                                      |
|-------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| V3          | independent            | Exact $\ell_p(n)$ against Monte Carlo design averages; correction identity; log-log slope $-2.06$ ; $n^2\Delta_n \in [20.16, 25.0]$                                                                                    |
| V3e,<br>V3f | independent            | Corollary 1 block structure $\mathbb{E}(\mathbf{S}^{-1}) = \text{blockdiag}(a_n, b_n \mathbf{I}_p)$ and the exact target formula for anisotropic and mean-shifted targets                                              |
| V4          | independent            | Loewner ordering (100 random precisions); misspecification monotonicity, crossing, and closed form against Monte Carlo                                                                                                 |
| V5          | mc                     | Shrinkage frontier: 36 of 72 interior cells worse than the better endpoint (max excess 0.221); grid-oracle sanity labelled tautological; EPPV monotone in $\lambda$ in every cell                                      |
| V6          | independent            | Leverage values $\{3, 1.5, 11.25, 5.5, 1.5\}$ ; noisy-source reversal 2.60 vs 2.30 (first order) and 2.70 vs 2.36 (exact, V6c)                                                                                         |
| V7          | independent            | $\rho$ -invariant allocation ranking; feasibility thresholds, first-order and exact                                                                                                                                    |
| V8          | independent            | Linear-oracle factor and the $pv/2$ maximal gain at $\eta = 1$                                                                                                                                                         |
| V9          | quadrature/mc          | JS+ strict dominance on the grid ( $p = 3, 5, 10$ ); Stein identity against the risk decomposition; origin risks $2v$ (plain) and $1.400v$ (JS+)                                                                       |
| V10         | quadrature             | EB boundary: violation exhibited at $p = 3$ ( $3.19v$ at $\eta = 4$ ); strict dominance at $p = 4$ over a declared non-centrality grid; origin display 0.5413                                                          |
| V11         | quadrature             | $p = 2$ divergence through capped inverse-norm moments growing without bound                                                                                                                                           |
| V12         | proof/independent      | Fixed-rule SURE unbiasedness; numerical SURE minimizer equals the EB factor; exact selected-rule optimism identity and direct Monte Carlo contrast                                                                     |
| V13         | quadrature/mc          | Full-matrix weighted quadrature against eigenbasis evaluation and direct MC, including a nondiagonal $\mathbf{W}$ ; dominance window, JS+/EB rule-specific thresholds, origin premium, and empty-window excesses       |
| V14         | independent/quadrature | Marginal HB calibration at $p = 2, 5, 10$ ; adaptive-integral diagnostics and interpolant accuracy; prior-scale sensitivity (0.386/0.712/0.897); log-uniform cutoff sensitivity documented (0.401 $\rightarrow$ 0.104) |
| V15         | mc                     | Canonical quadrature against 200,000-rep Monte Carlo for the menu                                                                                                                                                      |

| Block | Evidence | Statement checked                                                                                |
|-------|----------|--------------------------------------------------------------------------------------------------|
| V16   | mc       | Coverage: nominal .950–.960; $\rho$ misassumption .713/.592 at 0.9; conservative below the truth |

### F.3. Claim-boundary checklist.

TABLE F.2. What the verification suite supports and what it does not. Block identifiers and evidence types are defined in this appendix.

| Claim                                                                                                                                                                                          | Status                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| Diffuse conditional EPPV equals the Tipton–Mamakos first-order benchmark                                                                                                                       | Supported (V2, transcription)                       |
| Exact correction $\sum_a \sigma_a^2 p(p+3)/\{n_a(n_a-p-2)\}$ , order $n^{-2}$ , positive for $p \geq 1$                                                                                        | Supported (V3, independent MC)                      |
| Exact cross-population risk $a_n + b_n(L_{T S} - 1)$ , Gaussian sources, arbitrary targets                                                                                                     | Supported (V3e, V3f, V6c)                           |
| Prior precision weakly reduces the EPPV estimation component                                                                                                                                   | Supported (V4)                                      |
| A misspecified prior mean can raise fixed-truth MSE above the diffuse analysis                                                                                                                 | Supported (V4)                                      |
| An interior shrinkage penalty can be worse than the better endpoint                                                                                                                            | Supported (V5; 36 of 72 cells)                      |
| Prior-matched EPPV is monotone in the penalty and cannot locate the oracle                                                                                                                     | Supported (V5)                                      |
| Relative leverage can fall below, equal, or exceed $q$ ; low leverage with noisy source outcomes can still raise total risk                                                                    | Supported (V6); anchor result, made computable here |
| Fixed-floor design rankings are $\rho$ -invariant; feasibility and coverage are $\rho$ -sensitive                                                                                              | Supported (V7, V16)                                 |
| Linear-oracle characterization; $pv/2$ maximal gain at $\eta = 1$                                                                                                                              | Supported (V8, quadrature)                          |
| JS+ dominance for $p \geq 3$ ; EB dominance for $p \geq 4$ and its failure at $p = 3$                                                                                                          | Proved; quadrature (V9, V10)                        |
| Target-weighted window and boundary; JS+ threshold $p_{\text{eff}} \geq (p+2)/2$ ; EB/SURE+ threshold $p_{\text{eff}} \geq (p+4)/2$ ; empty Euclidean-factor window at $p_{\text{eff}} \leq 2$ | Bock–Berger specialization; quadrature/MC (V13)     |
| No rule is weakly better than both endpoints for all truths                                                                                                                                    | Impossible (Theorem 5)                              |
| SURE is unbiased at every fixed penalty; its minimizer is the EB rule                                                                                                                          | Proved; V12                                         |

| Claim                                                                                           | Status                                                                                         |
|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Minimized SURE is an unbiased estimate of the selected rule’s risk                              | False; exact positive optimism correction (Prop. 5(iii), V12)                                  |
| HB marginal calibration and risk sensitivity to halving or doubling the elicited prior scale    | Supported (V14); sensitivity is material, and the replaced log-uniform prior was cutoff-driven |
| Canonical theory describes finite designs at a stated accuracy in every cell                    | Reported cell by cell (V15, Sec. 7.3); no single adequacy claim                                |
| General Bayesian dominance of the frequentist calculus                                          | Not claimed                                                                                    |
| Optimality of any menu rule within all data-driven rules                                        | Not claimed                                                                                    |
| Guarantees under estimated scale, unbalanced leverage, coefficient drift, or nonlinear learners | Not claimed; finite-design track quantifies, theory open                                       |

TABLE F.3. Figure-level settings (suite 02\_simulation.R).

| Display             | Factors                                                                                                                                                             | Replications            |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Fig. 2              | $n \in \{20, \dots, 500\}$ ; exact vs first order vs MC                                                                                                             | 1,500 designs per $n$   |
| Fig. B.1            | $n_0 \in \{0, \dots, 300\}$ ; $n \in \{20, 50, 100\}$ ; offsets $\{0, .25, .5\}\sigma$ at $n = 50$                                                                  | 300 designs per cell    |
| Fig. 3              | $\lambda \in \{0, 0.5, 1, 2, 4, 8, 16, 32, 64, \infty\}$ ; three signal scenarios; $n \in \{20, 50, 200\}$                                                          | 300 per cell            |
| Fig. 4              | five source-target scenarios; first-order and exact totals at $n = 20$                                                                                              | exact + 20,000 MC check |
| Fig. 5              | $\rho \in [-0.5, 0.9]$ ; $R^* = 2.25$ ; first-order and exact curves                                                                                                | exact                   |
| Fig. D.1            | $n \in \{20, 100\}$ ; $n_0 \in \{0, 5, 20, 50\}$ ; offsets $\{0, .5\}\sigma$ ; $\rho_{\text{assumed}} \in \{-.5, 0, .3, .6, .9\}$ , $\rho_{\text{true}} = .3$       | 4,000 per cell          |
| Figs. 6, 7, and F.1 | $p \in \{2, 5, 10\}$ ; $\eta$ on the 13-point grid $[0, 8]$ ; full menu                                                                                             | quadrature + 200,000 MC |
| Fig. 8(b)           | three $\mathbf{W}$ shapes at $p = 6$ ; $b \in [0.1, 8]$ ; signal probe set along the top axis                                                                       | quadrature              |
| Figs. 9 and F.3     | $n \in \{20, 50, 100, 200\}$ , $p = 5$ , $\eta \in \{0, .25, 1, 4\}$ ; estimated scale; heteroscedastic (1,2) at $n \in \{20, 100\}$ ; CV five folds, 12-point grid | 2,000 (800 CV) per cell |
| Fig. 10             | preposterior scenarios skeptic/agnostic/opti-mist over the $\eta$ grid                                                                                              | quadrature              |
| Fig. F.2            | HB prior scale $\times \{0.5, 1, 2\}$ over the $\eta$ grid                                                                                                          | quadrature              |

**F.4. Simulation settings.** Seeds: verification 20260801 (block-local seeds 20260802 through 20260812 where noted in the script); canonical Monte Carlo 20260813; finite designs 20260814; cross-validation 20260815; heteroscedastic 20260816; bridge

and prior tables 20260817/20260818. Software: R 4.6.0; no Markov chain Monte Carlo anywhere in the suite. The suite writes `verification_report.md` and `key_numbers.csv`, exits with a nonzero status on any failure, and the shipped run reports zero failures.

**F.5. Finite-design secondary tracks.** The complete homoscedastic comparison contains twelve tuned-rule cells at each arm size: JS+, EB/SURE, and hierarchical Bayes at  $\eta \in \{0, .25, 1, 4\}$ . Relative deviations from canonical risk range from 23.2% to 84.6% at  $n = 20$ , 7.3% to 17.8% at  $n = 50$ ,  $-0.4\%$  to  $8.5\%$  at  $n = 100$ , and  $-0.8\%$  to  $5.0\%$  at  $n = 200$ . Tuned-rule ordering matches the canonical ordering in every cell at  $n \geq 100$ ; it differs at  $\eta = 4$  for  $n = 20$  and  $50$ .

Under heteroscedastic arm scales  $(1, 2)$  with the common-scale fit retained, tuned-rule risks range from 0.3% below to 9.8% above their homoscedastic counterparts at  $n = 20$ , and from 0.4% to 4.1% above at  $n = 100$ . At  $\eta = 1$ , for example, the hierarchical rule is  $3.799v$  rather than  $3.637v$  at  $n = 20$  and  $3.047v$  rather than  $3.018v$  at  $n = 100$ . Five-fold CV pays zero-signal risks  $1.245v$ ,  $0.811v$ ,  $0.828v$ , and  $0.613v$  at  $n = 20, 50, 100, 200$ . It lies below the realized CATE endpoint in 15 of 16 cells; at  $n = 200$ ,  $\eta = 4$ , the values are  $5.165v$  and  $5.157v$ , a difference of  $0.008v$  that is small relative to the reported Monte Carlo standard error. These are descriptive finite-design comparisons, not extensions of the canonical guarantees.

F.6. **Supplementary figures.** Three supplementary displays complete the record: the dimension panel behind the main canonical summary, the hierarchical-Bayes prior-scale sensitivity behind block V14, and the heteroscedastic finite-design panel.

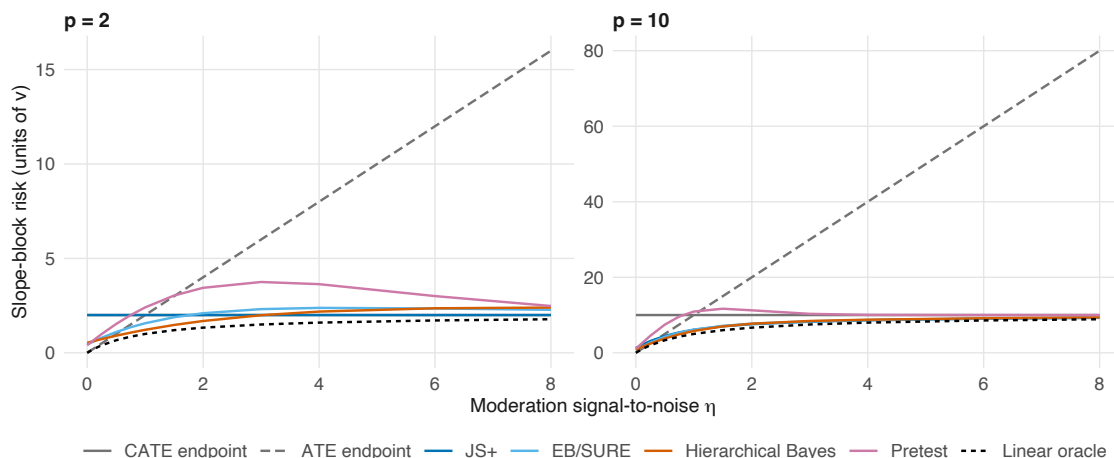


FIGURE F.1. The rule menu at  $p = 2$  and  $p = 10$  (canonical layer, quadrature curves). At  $p = 2$  the Stein window is empty, the James–Stein rule coincides with the CATE endpoint, and no rule is uniformly safe (Theorem 3, part iv); the empirical-Bayes and hierarchical rules still perform well on the grid, but without a guarantee. At  $p = 10$  the guarantees bind more tightly.

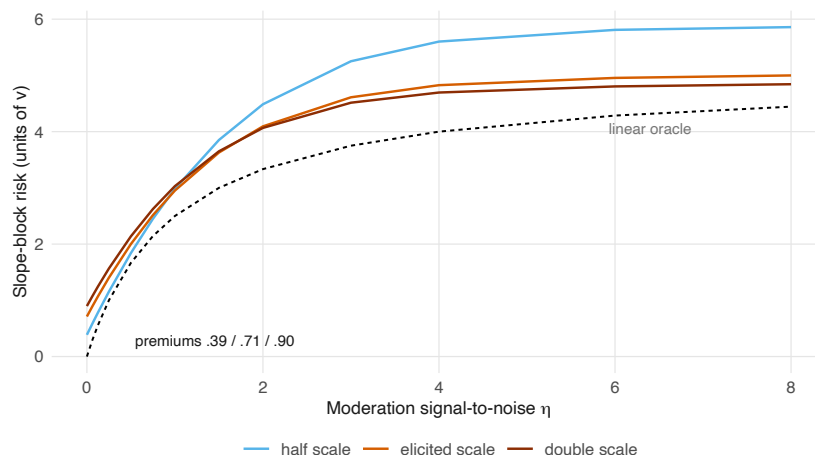


FIGURE F.2. Hierarchical-Bayes risk under half, the marginally calibrated, and twice the elicited prior scale ( $p = 5$ ; the dotted line is the linear oracle). The zero-signal premiums are  $0.386v$ ,  $0.712v$ , and  $0.897v$ , a max/min ratio of 2.32, so the scale sensitivity is material.

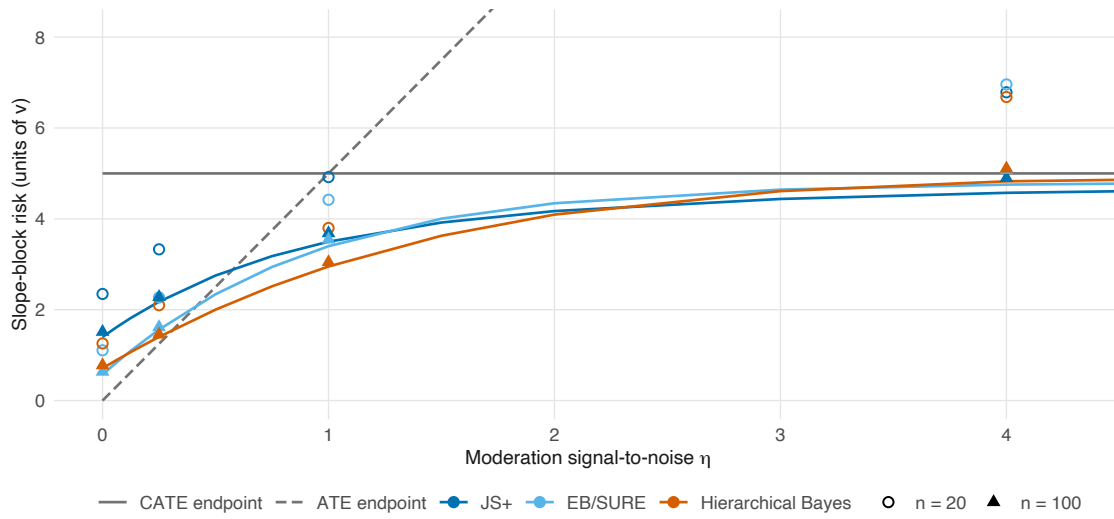


FIGURE F.3. Heteroscedastic finite-design validation at  $p = 5$  (points; 2,000 replications per cell) against canonical quadrature curves (lines). The arms have residual scales (1, 2) while the rules retain the common-scale working convention.